

314458: Laboratory Practice –II (Cloud Computing)

Third Year – Information Technology
(2019 Course)

Academic Year 2025-26 Semester – II

Teaching Scheme:	Credit Scheme:	Examination Scheme:	
Practical (PR): 04 hrs/week	02 Credit	PR: 25 Marks	TW: 50Marks



LABORATORY MANUAL V 1.0

KJ's Educational Institute

TRINITY ACADEMY OF ENGINEERING, PUNE

(Approved by AICTE, New Delhi, Govt. of Maharashtra & affiliated to SPPU, DTE Code: EN6634)

(Accredited by NAAC with 'A+' Grade)

2025-2026

VISION

To provide excellent Information Technology education by building strong teaching and research environment.

MISSION

- 1) To transform the students into innovative, competent, and high-quality IT professionals to meet the growing global challenges.
- 2) To achieve and impart quality education with an emphasis on practical skills and social relevance.
- 3) To endeavor for continuous up-gradation of the technical expertise of students to cater to the needs of society.
- 4) To achieve an effective interaction with the industry for mutual benefits.

PROGRAM EDUCATIONAL OBJECTIVES

The students of the Information Technology course after passing out will

Sr. No.	Description
PEO1	Possess strong fundamental concepts in mathematics, science, engineering, and Technology to address technological challenges.
PEO2	Possess knowledge and skills in the field of Computer Science and Information Technology for analyzing, designing, and implementing complex engineering problems of any domain with innovative approaches.
PEO3	Possess an attitude and aptitude for research, entrepreneurship, and higher studies in the field of Computer Science and Information Technology.
PEO4	Have a commitment to ethical practices, societal contributions through communities, and life-long learning.
PEO5	Possess better communication, presentation, time management, and team work skills leading to responsible & competent professionals and will be able to address challenges in the field of IT at the global level.

PROGRAM OUTCOMES

The students in the Information Technology course are expected to know and be able to:

Sr. No.	PO's	Description
PO1	Engineering knowledge	An ability to apply knowledge of mathematics, computing, science, engineering and technology.
PO2	Problem analysis	An ability to define a problem and provide a systematic solution with the help of conducting experiments, analyzing the problem and interpreting the data.
PO3	Design/Development of Solutions	An ability to design, implement, and evaluate software or a software/hardware system, component, or process to meet desired need switch in realistic constraints.
PO4	Conduct Investigation of Complex Problems	An ability to identify, formulate, and provide essay schematic solutions to complex engineering /Technology problems.
PO5	Modern Tool Usage	An ability to use the techniques, skills, and modern engineering technology tools, and standard processes necessary for practice as an IT professional.
PO6	The Engineer and Society	An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems with necessary constraints and assumptions.
PO7	Environment and Sustainability	An ability to analyze and provide solutions for the local and global impact of information technology on individuals, organizations, and society.
PO8	Ethics	An ability to understand professional, ethical, legal, security and social issues and responsibilities.
PO9	Individual and Team Work	An ability to function effectively as an individual or a sate am member to accomplish a desired goal(s).
PO10	Communication Skills	An ability to engage in life-long learning and continuing professional development to cope up with fast changes in the technologies /tools with the help of electives, profession along animations and extra- curricular activities.
PO11	Project Management and Finance	An ability to communicate effectively in engineering community at large by means of effective presentations, report writing, paper publications, demonstrations.
PO12	Life-long Learning	An ability to understand engineering, management, financial aspects, performance, optimizations and time complexity necessary for professional practice.

DOCUMENT CONTROL

Reference Code	TAE-IT / Lab Manual Procedures
Version No	1.0
Compliance Status	Complete
Date of Compliance	24-12-2025
Security Classification	Department Specific
Document Status	Definitive
Review Period	Yearly

	Author
Signature	
Name	Mr. A. G. Rao
Designation	Assistant Professor

SYLLABUS

Savitribai Phule Pune University, Pune Third Year Information Technology (2019 Course) 314458: Laboratory Practice-II (Cloud Computing)			
Teaching Scheme:	Credit Scheme:	Examination Scheme:	
Practical (PR) : 04 hrs/week	02 Credit	PR :25 Marks	TW : 50Marks
Prerequisite Courses:			
- Basics of Computer Networks - Operating Systems			
Course Objectives:			
1. To develop web applications in cloud.			
2. To learn the design and development process involved in creating a cloud based application.			
Course Outcomes:			
On completion of the course, students will be able to			
CO1: To design and develop cloud based applications.			
CO2: To Simulate a cloud scenario using CloudSim.			
CO3: To design and deploy web applications in cloud environment.			
LIST OF ASSIGNMENTS			
1. Install Google App Engine. Create hello world app and other simple web applications using python/java.			
2. Use GAE launcher to launch the web applications.			
3. Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.			
4. Find a procedure to transfer the files from one virtual machine to another virtual machine.			
5. Find a procedure to launch virtual machine using trystack (Online Openstack Demo Version)			
6. Design and deploy a web application in a PaaS environment.			
7. Design and develop custom Application (Mini Project) using Salesforce Cloud.			
8. Design an Assignment to retrieve, verify, and store user credentials using Firebase Authentication, the Google App Engine standard environment, and Google Cloud Data store			
CASE STUDIES			
- Data storage security in private cloud - Application of IoT/Ubiquitous based on cloud			

• Tools for building private cloud
Text Books:
1. Thomas Erl, Zaigham Mahmood and Ricardo Puttini, Cloud Computing: Concepts, Technology & Architecture, Pearson, ISBN :978 9332535923, 9332535922, 1 st Edition
2. Anthony T. Velte Toby J. Velte, Robert Elsenpeter, "Cloud Computing: A Practical Approach" 2010, The McGraw-Hill.
Reference Books:
1. Rajkumar Buyya, Christian Vecchiola, S. ThamaraiS elvi, Mastering Cloud Computing: Foundations and Applications Programming, McGraw Hill, ISBN: 978 1259029950, 1259029956.
2. Gautam Shrof, "ENTERPRISE CLOUD COMPUTING" Technology Architecture, Applications, Cambridge University Press, ISBN: 9780511778476
3. Srinivasan, J. Suresh, Cloud Computing: A practical approach for learning and implementation, Pearson, ISBN :9788131776513.
4. Jack J. Dongarra, Kai Hwang, Geoffrey C. Fox, Distributed and Cloud Computing: From Parallel Processing to the Internet of Things, Elsevier, ISBN :9789381269237, 9381269238, 1st Edition.
5. Brian J.S. Chee and Curtis Franklin, Jr., Cloud Computing: Technologies and Strategies of the Ubiquitous Data Center, CRC Press, ISBN :9781439806128.
6. Kris Jamsa, Cloud Computing: Saas, Paas, Iaas, Virtualization, Business Models, Mobile, Security, and More, Jones and Bartlett, ISBN :9789380853772.
7. John W. Ritting house, James F. Ransome, Cloud Computing Implementation, Management, and Security, CRC Press, ISBN : 978 1439806807, 1439806802.
8. Karl Matthias, Sean P. Kane, Docker: Up and Running, OReilly, ISBN: 9781491917572, 1491917571.
9. Barrie Sosinsky, Cloud Computing Bible, Wiley, ISBN: 978 8126529803.
10. Ronald L. Krutz and Russell D. Vines, Cloud Security: A Comprehensive guide to Secure Cloud Computing, Wiley, ISBN: 9788126528097.
11. Scott Adkins, John Belamaric, Vincent Giersch, Denys Makogon, Jason E. Robinson, OpenStack: Cloud Application Development, Wrox, ISBN :9781119194316.
12. Kailash Jayaswal, Jagannath Kallakurchi, Donald J. Houde, Cloud Computing Black Book ,Wiley Dreamtech,ISBN:9789351194187



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2025-2026

ACADEMIC YEAR 2025-26, SEMESTER-II

Name of Student	:		PRN No.	
Student Roll No.	:		Class	Third Year
Subject	:	Laboratory Practice-II (Cloud Computing)	Batch	

I N D E X

Sr.no	Date	Name of Experiment	Page No.
1		Comparative Study of Open-Source Data Analysis tools.	
2		Use GAE launcher to launch the web applications.	
3		Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.	
4		Find a procedure to transfer the files from one virtual machine to another virtual machine.	
5		Find a procedure to launch a virtual machine using trystack (Online Openstack Demo Version)	
6		Design and deploy a web application in a PaaS environment.	
7		Design and develop custom Applications (Mini Project) using Salesforce Cloud.	
8		Design an Assignment to retrieve, verify, and store user credentials using Firebase Authentication, the Google App Engine standard environment, and the Google Cloud Data store.	

SCHEDULE

Sr. No.	Name of Experiment	No. of Hrs.	Week
1	Install Google App Engine. Create hello world app and other simple web applications using python/java.		
2	Use GAE launcher to launch the web applications.		
3	Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.		
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Assignment No. 1

Title:

Install Google App Engine. Create hello world app and other simple web applications using python/java.

Objectives:

1. To study/Install GAE Environment
2. To learn GAE simple coding
3. To create and run a simple hello world web application on Google Cloud SDK Shell

Theory:**What is Google App Engine?**

Google App Engine (GAE) is a service (platform-as-a-service) product, a scalable environment, and a cloud computing technology that Google provides for developing/building and hosting Web applications in Google's data centers. GAE is a scalable runtime environment. These dynamic scales as demand change over time because of Google's vast computing infrastructure. GAE offers a secure execution environment and a collection of services, that makes it easier to develop scalable and high-performance Web apps.

Web applications hosted on GAE are sandboxed and run across multiple servers for redundancy and allow for the scaling of resources according to the traffic requirements of the moment. App Engine automatically allocates additional resources to the servers to accommodate the increased load.

GAE requires these applications to be written in one of a few supported languages, namely: Java, Python, PHP, and Go. It stores data in Google Bigtable and uses the Google query language.

GAE takes advantage of Google's vast & efficient computing infrastructure to scale dynamically with changing demands over time. "Effortless" comes to mind when we talk of GAE as it frees developers from administrator work of managing infrastructure and provides effortless development of applications. The GAE SDK facilitates the testing and professionalization of applications by emulating the production runtime environment and allowing developers to design and test applications on their own PCs.

The App Engine architecture

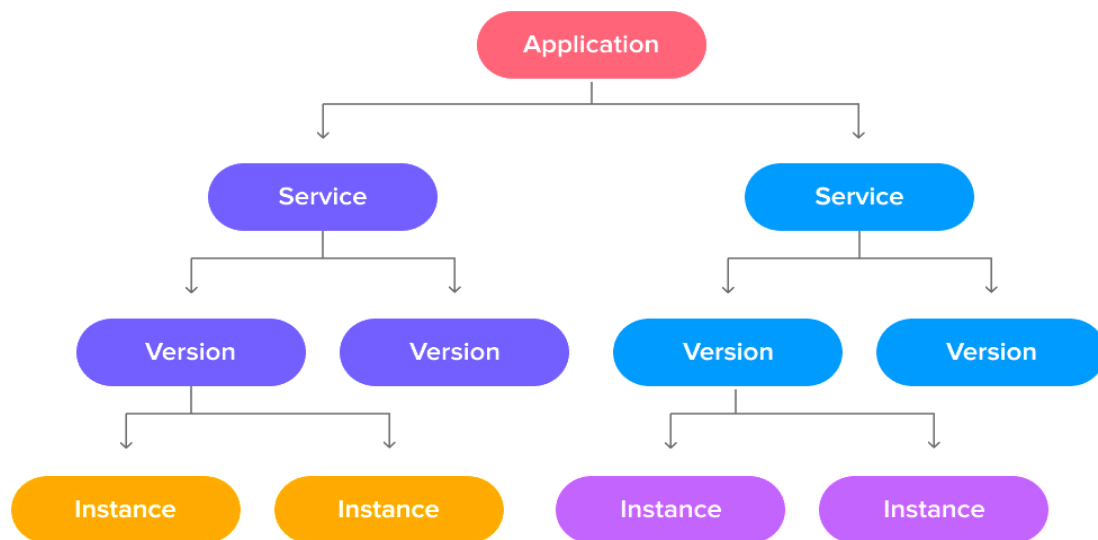


Figure 1: Google App Engine Architecture

Services provided by App Engine include:

- Platform as a Service (PaaS) to build and deploy scalable applications
- Hosting facility in fully-managed data centers
- A fully-managed, flexible environment platform for managing application server and infrastructure
- Support in the form of popular development languages and developer tools

How is GAE used?

GAE is a fully managed, serverless platform that is used to host, build and deploy web applications. Users can create a GAE account, set up a software development kit and write application source code. They can then use GAE to test and deploy the code in the cloud.

One way to use GAE is building scalable mobile application back ends that adapt to workloads as needed. Application testing is another way to use GAE. Users can route traffic to different application versions to A/B test them and see which version performs better under various workloads.

Major Features of Google App Engine

These prominent features enable GAE to be a hassle-free service platform.

- 1) **API selection:** GAE has several built-in APIs, including the following five:
 - Blobstore for serving large data objects;
 - GAE Cloud Storage for storing data objects;
 - Page Speed Service for automatically speeding up webpage load times;
 - URL Fetch Service to issue HTTP requests and receive responses for efficiency and scaling;

- Memcache for a fully managed in-memory data store.
- 2) **Managed infrastructure:** Google manages the back-end infrastructure for users. This approach makes GAE a serverless platform and simplifies API management. The engine ensures that your web apps are secure and running and saves them from malware and threats by enabling the firewall.
 - 3) **Collection of Several development languages:** GAE supports a number of languages, including GO, PHP, Java, Python, NodeJS, .NET and Ruby. It also supports custom runtimes. Every language has its supporting SDK and runtime.
 - 4) **Support for legacy runtimes:** GAE supports legacy runtimes, which are versions of programming languages no longer maintained. Examples include Python 2.7, Java 8 and Go 1.11.
 - 5) **Effective Diagnostic & Security Services:** Cloud Monitoring and Cloud Logging help run app scans to identify bugs. Further, app reporting documents help developers fix bugs on an immediate basis. GAE also sets up rules for access, SSL certificates on custom domains, and TSL certificates without any extra charge to create user firewall systems.
 - 6) **Security features:** GAE enables users to define access policies with the GAE firewall and managed Secure Sockets Layer/Transport Layer Security certificates for free.
 - 7) **Traffic splitting:** The app engine automatically routes the incoming traffic to different versions of the apps as a part of A/B testing.
 - 8) **Versioning:** Applications in Google App Engine function as a set of micro services that refer back to the main source code. Every time code is deployed to a service with the corresponding GAE configuration files, a version of that service is created.

Benefits of Google App Engine?



Figure 2: Benefits of GAE

Install and configure Google App Engine

Pre-Requisites:

1) Python 2.5.4

Before you get started, we need to install Python. If you don't already have Python 2.5.4 installed in your computer, download and Install Python 2.5.4 from:
(The version you download will depend on whether or not your PC is 32 or 64bit).

<https://www.python.org/downloads/release/python-254/>

2) Download and Install

You can download Google App Engine SDK by going to:

<https://cloud.google.com/appengine/downloads?csw=1>

and download the appropriate install package

Download the Google App Engine SDK

Before downloading, please read the [Terms](#) that govern your use of the App Engine SDK.

Please note: The App Engine SDK is under **active development**, please keep this in mind as you explore its capabilities. See the [SDK Release Notes](#) for the information on the most recent changes to the App Engine SDK. If you discover any issues, please feel free to notify us via our [Issue Tracker](#).

Platform	Version	Package	Size	SHA1 Checksum
Windows	1.1.5 - 10/03/08	GoogleAppEngine_1.1.5.msi	2.5 MB	e974312b4aefc0b3873ff0d93eb4c525d5e88c30
Mac OS X	1.1.5 - 10/03/08	GoogleAppEngineLauncher-1.1.5.dmg	3.6 MB	f62208ac01c1b3e39796e58100d5f1b2f052d3e7
Linux/Other Platforms	1.1.5 - 10/03/08	google_appengine_1.1.5.zip	2.6 MB	cbb9ce817bdabf1c4f181d9544864e55ee253de1

Download the Windows installer the simplest thing is to download it to your Desktop or another folder that you remember.

Download the archive file best suited to your operating system. Most machines will run the 64-bit package. If you'd like to check, run `uname -m` to verify if you're running a 64-bit system.

Platform	Package	Size	SHA256 Checksum
Linux 64-bit (x86_64)	google-cloud-sdk229.0.0-linuxx86_64.tar.gz	25.6 MB	b1c87fc9451598a76cf66978dd8aa06482bfced639b56cf31559dc2c7f8b7b90
Linux 32-bit (x86)	google-cloud-sdk229.0.0-linuxx86.tar.gz	25.2 MB	ee8c45f8018d0fee92b07c32cc6d8c891241da0b88bfe289d4e58e6746c3f668

Installation of Google App Engine

The Google Cloud CLI works on Windows 8.1 and later and Windows Server 2012 and later.

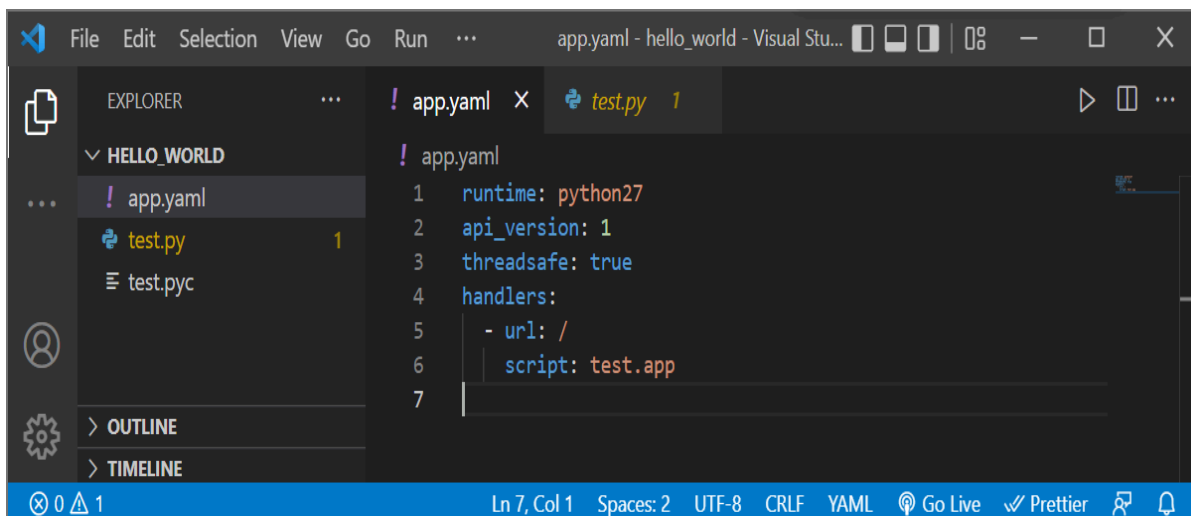
- 1) Download the Google Cloud CLI installer from the following link - <https://dl.google.com/dl/cloudsdk/channels/rapid/GoogleCloudSDKInstaller.exe>
- 2) Launch the installer and follow the prompts. The installer is signed by Google LLC. If you're using a screen reader, check the Turn on screen reader mode checkbox. This option configures gcloud to use status trackers instead of unicode spinners, display progress as a percentage, and flatten tables. For more information, see the Accessibility features guide.
- 3) Cloud SDK requires Python; supported versions are Python 3 (preferred, 3.5 to 3.8) and Python 2 (2.7.9 or later). By default, the Windows version of Cloud SDK comes bundled with Python 3 and Python 2. To use Cloud SDK, your operating system must be able to run a supported version of Python. The installer installs all necessary dependencies, including the needed Python version. While Cloud SDK installs and manages Python 3 by default, you can use an existing Python installation if necessary by unchecking the option to Install Bundled Python. See gcloud topic startup to learn how to use an existing Python installation.
- 4) After installation is complete, the installer gives you the option to create Start Menu and Desktop shortcuts, start the Google Cloud CLI shell, and configure the gcloud CLI. Make sure that you leave the options to start the shell and configure your installation selected. The installer starts a terminal window and runs the gcloud init command.
- 5) The default installation doesn't include the App Engine extensions required to deploy an application using gcloud commands. These components can be installed using the gcloud CLI component manager.

Troubleshooting tips:

- 1) If your installation is unsuccessful due to the find command not being recognized, ensure your PATH environment variable is set to include the folder containing find. Usually, this is C:\WINDOWS\system32;.
- 2) If you uninstalled the gcloud CLI, you must reboot your system before installing the gcloud CLI again.
- 3) If unzipping fails, run the installer as an administrator.

Source Code

File 1: app.yaml

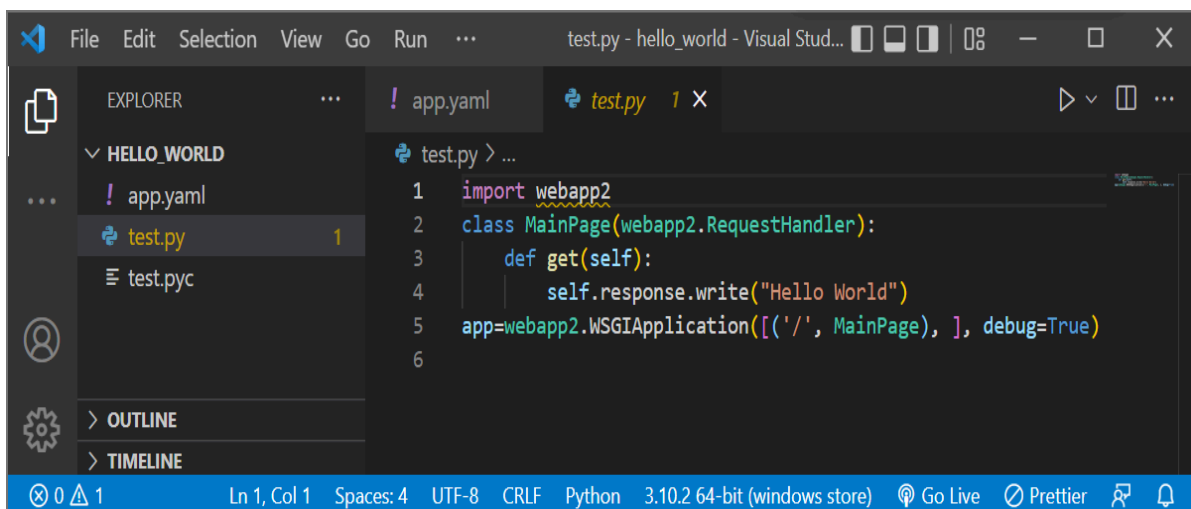


The screenshot shows the Visual Studio Code interface with the 'HELLO_WORLD' project open. The Explorer sidebar on the left shows the file structure with 'app.yaml' and 'test.py' (1) under 'HELLO_WORLD'. The main editor area displays the content of 'app.yaml'.

```
app.yaml
1 runtime: python27
2 api_version: 1
3 threadsafe: true
4 handlers:
5   - url: /
6     script: test.app
7
```

The status bar at the bottom indicates 'Ln 7, Col 1', 'Spaces: 2', 'UTF-8', 'CRLF', 'YAML', and other settings.

File 2: test.py



The screenshot shows the Visual Studio Code interface with the 'HELLO_WORLD' project open. The Explorer sidebar on the left shows the file structure with 'app.yaml' and 'test.py' (1) under 'HELLO_WORLD'. The main editor area displays the content of 'test.py'.

```
test.py > ...
1 import webapp2
2 class MainPage(webapp2.RequestHandler):
3     def get(self):
4         self.response.write("Hello World")
5 app=webapp2.WSGIApplication(['/', MainPage], debug=True)
6
```

The status bar at the bottom indicates 'Ln 1, Col 1', 'Spaces: 4', 'UTF-8', 'CRLF', 'Python', '3.10.2 64-bit (windows store)', and other settings.

How to Run Program (Linux based):

Now as we have finished installing app engine, now it's time to create and upload an app. In this case, we will be taking an example of a "HELLO WORLD" app in python.

1. As we already have made sure that we have python installed in our system, It will be easier for us to clone existing code and deploy it rather than creating our own so we will use a python docs sample. Run the command
"git clone https://github.com/GoogleCloudPlatform/python-docs-samples".
2. cd python docs- samples/appengine/standard/hello_world
3. dev_appserver.py app.yaml

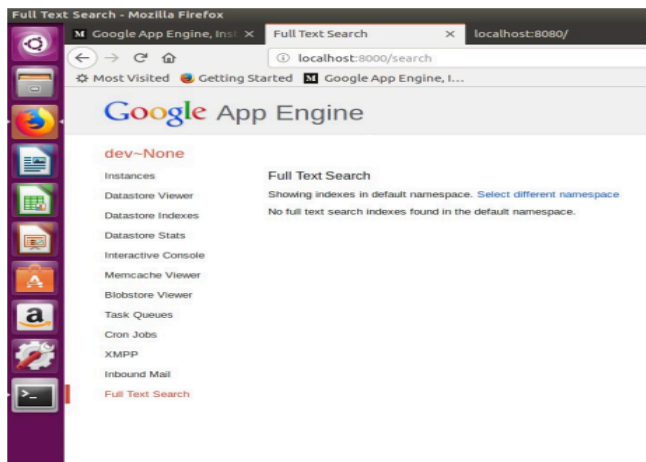


Figure 3: Launch of GAE

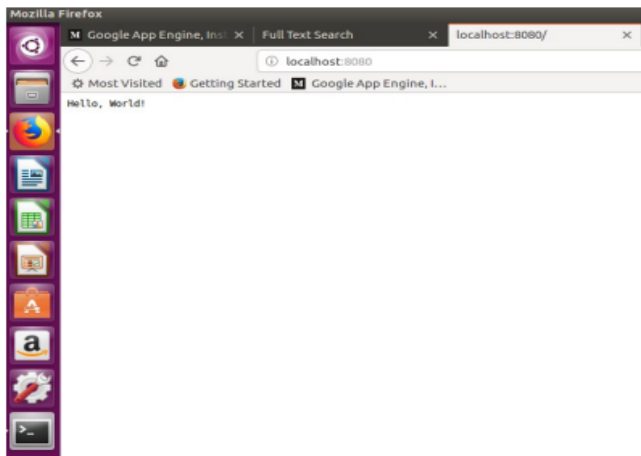
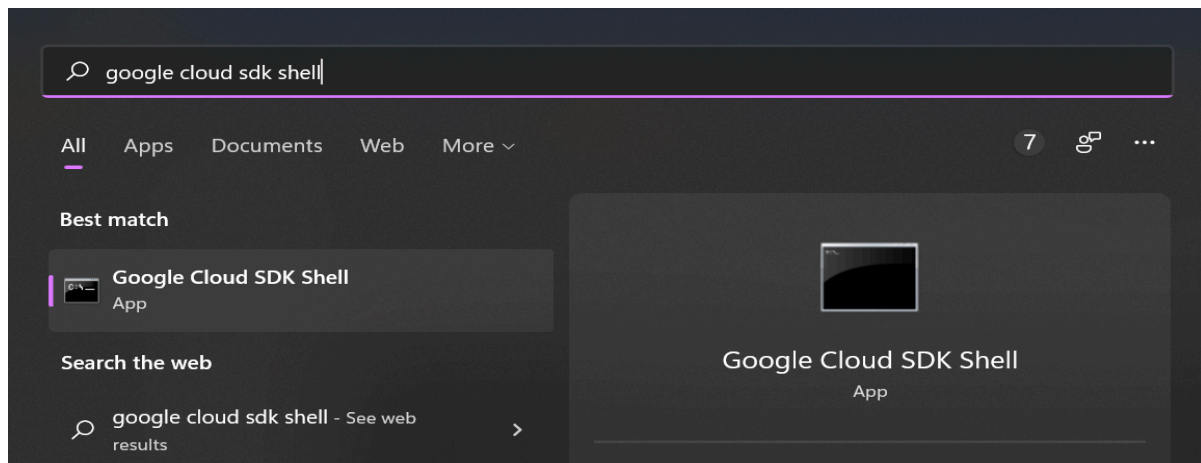


Figure 4: GAE App Output check

Step-by-step procedure to launch the hello world app on Google App Engine

Step 1 – Open 'Google Cloud SDK Shell'



Step 2 – Run the following command

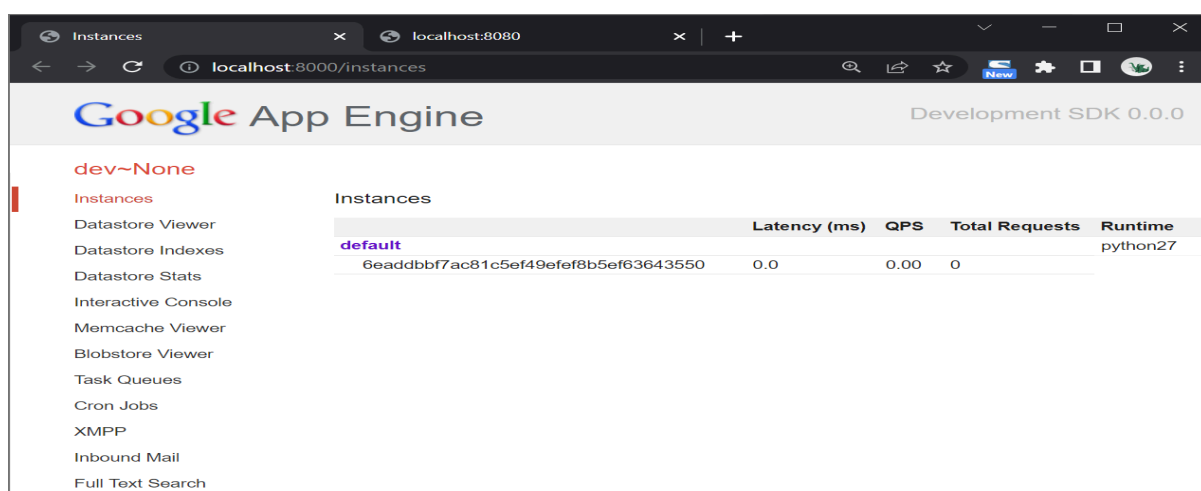
```

C:\Users\lenovo\AppData\Local\Google\Cloud SDK>py google-cloud-sdk/bin/dev_appserver.py C:\Users\lenovo\Desktop\hello_world

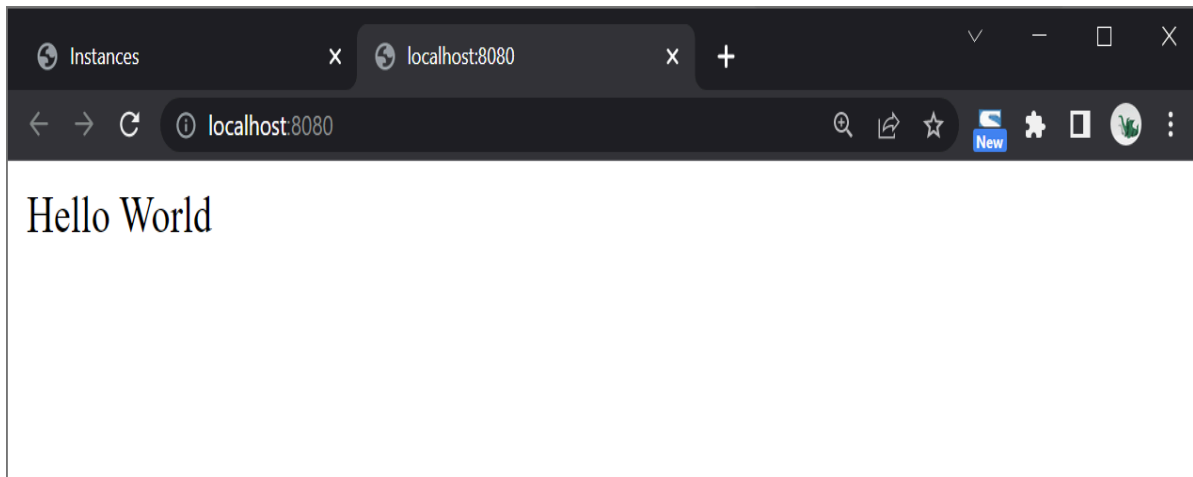
Welcome to the Google Cloud CLI! Run "gcloud -h" to get the list of available commands.
---

INFO     2022-04-07 15:06:12,234 devappserver2.py:316] Skipping SDK update check.
WARNING  2022-04-07 15:06:13,285 simple_search_stub.py:1196] Could not read search indexes from c:\users\lenovo\appdata\local\temp\appengine.None\search_indexes
INFO     2022-04-07 15:06:13,289 <string>:384] Starting API server at: http://localhost:50168
INFO     2022-04-07 15:06:13,296 dispatcher.py:281] Starting module "default" running at: http://localhost:8080
INFO     2022-04-07 15:06:13,301 admin_server.py:150] Starting admin server at: http://localhost:8000
  
```

Step 3 – Now go to <http://localhost:8000> in your browser. You will see the following dashboard,



Step 4 – Click on the default button. You will be redirected to <http://localhost:8080>

**Conclusion:**

Thus following the step-by-step procedure of installation and execution of a simple app, we have successfully created and run a simple hello world web application on Google App Engine.

Assignment No. 2

Title :

Use GAE launcher to launch the web applications.

Objective:

4. To learn GAE Environment and launcher.
5. To learn the design and deployment process involved in creating cloud-based applications.

Theory:

What is Google App Engine?

Google App Engine (GAE) is a service (platform-as-a-service) product, a scalable environment, and a cloud computing technology that Google provides for developing/building and hosting Web applications in Google's data centers. GAE is a scalable runtime environment. These dynamic scales as demand change over time because of Google's vast computing infrastructure. GAE offers a secure execution environment and a collection of services, that makes it easier to develop scalable and high-performance Web apps.

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Install and configure Google App Engine

Pre-Requisites:

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(The version you download will depend on whether or not your PC is 32 or 64bit).

<https://www.python.org/downloads/release/python-254/>

2) Download and Install

You can download Google App Engine SDK by going to:

<https://cloud.google.com/appengine/downloads?csw=1>

and download the appropriate install package

Download the Google App Engine SDK

Before downloading, please read the [Terms](#) that govern your use of the App Engine SDK.

Please note: The App Engine SDK is under **active development**, please keep this in mind as you explore its capabilities. See the [SDK Release Notes](#) for the information on the most recent changes to the App Engine SDK. If you discover any issues, please feel free to notify us via our [Issue Tracker](#).

Platform	Version	Package	Size	SHA1 Checksum
Windows	1.1.5 - 10/03/08	GoogleAppEngine_1.1.5.msi	2.5 MB	e974312b4aefc0b3873ff0d93eb4c525d5e88c30
Mac OS X	1.1.5 - 10/03/08	GoogleAppEngineLauncher-1.1.5.dmg	3.6 MB	f62208ac01c1b3e39796e58100d5f1b2f052d3e7
Linux/Other Platforms	1.1.5 - 10/03/08	google_appengine_1.1.5.zip	2.6 MB	cbb9ce817bdabf1c4f181d9544864e55ee253de1

Download the Windows installer the simplest thing is to download it to your Desktop or another folder that you remember.

Download the archive file best suited to your operating system. Most machines will run the 64-bit package. If you'd like to check, run `uname -m` to verify if you're running a 64-bit system.

Platform	Package	Size	SHA256 Checksum
Linux 64-bit (x86_64)	google-cloud-sdk229.0.0-linuxx86_64.tar.gz	25.6 MB	b1c87fc9451598a76cf66978dd8aa06482bfced639b56cf31559dc2c7f8b7b90
Linux 32-bit (x86)	google-cloud-sdk229.0.0-linuxx86.tar.gz	25.2 MB	ee8c45f8018d0fee92b07c32cc6d8c891241da0b88bfe289d4e58e6746c3f668

Installation of Google App Engine

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- 1) Download the Google Cloud CLI installer from the following link - <https://dl.google.com/dl/cloudsdk/channels/rapid/GoogleCloudSDKInstaller.exe>
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Troubleshooting tips:

- 1) If your installation is unsuccessful due to the find command not being recognized, ensure your PATH environment variable is set to include the folder containing find. Usually, this is C:\WINDOWS\system32;.
- 2) If you uninstalled the gcloud CLI, you must reboot your system before installing the gcloud CLI again.
- 3) If unzipping fails, run the installer as an administrator.

Source Code

Step-by-step procedure to launch the Web Application on Google App Engine

Step 1 – Make a folder for your Google App Engine applications. For example, make the on Desktop called “cc-ass2” – the path to this folder is:

`C:\Users\cheta\Desktop\cc-ass2`

Step 2 – Make a sub--folder in within cc-ass2 called “www” – the path to this folder would be:

`C:\Users\cheta\Desktop\cc-ass2\www`

Step 3 – Using a text editor such as Visual Studio Code (<https://code.visualstudio.com/>), create a file called app.yaml in the cc-ass2 folder with the following contents:

```
runtime: python27
api_version: 1
threadsafe: true

handlers:
- url: /
  static_files: www/index.html
  upload: www/index.html

- url: /(.*)
  static_files: www/\1
  upload: www/(.*)
```

Note: Please do not copy and paste these lines into your text editor – you might end up with strange characters – simply type them into your editor.

Step 4 – Create a file in the www folder called “index.html” with few lines of code in it:

```
! app.yaml  index.html X
www > <> index.html > html > style
1  <!DOCTYPE html>
2  <html lang="en">
3
4  <head>
5      <meta charset="UTF-8">
6      <meta http-equiv="X-UA-Compatible" content="IE=edge">
7      <meta name="viewport" content="width=device-width, initial-scale=1.0">
8      <title>Fitness</title>
9  </head>
10 <link rel="stylesheet" href="css/style.css">
11 > <style>...
135 </style>
136
137 <body>
138 > <header class="header">...
162 </header>
163 <div class="container">
164     <h1>Welcome to he Fitness gym</h1>
165     <form action="/" method="post" class="basicData">
166         <div class="form-group">
167             <input type="text" name="Name" placeholder="Enter your Name">
168         </div>
169         <div class="form-group">
170             <input type="text" name="Age" placeholder="Enter your Age">
171         </div>
172         <div class="form-group">
173             <input type="text" name="Gender" placeholder="Enter your Gender">
174         </div>
175         <div class="form-group">
176             <input type="number" name="contact" placeholder="Enter your contact no. ">
177         </div>
178         <div class="form-group">
179             <input type="text number" name="Address" placeholder="Enter your Address">
180         </div>
181         <div class="form-group">
182             <input type="text" name="City" placeholder="Enter your City">
183         </div>
184         <button class="btn">submit</button>
```

Step 5 – Start the Google cloud sdk shell program that can be found under Applications.

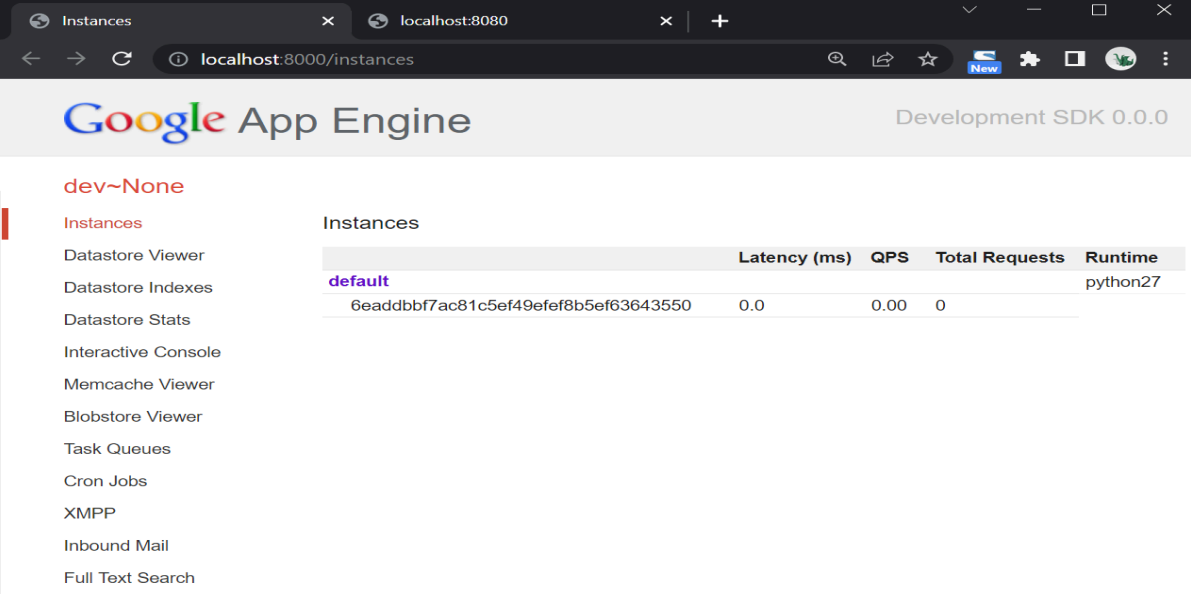
>py google-cloud-sdk\bin\dev_appserver.py
C:\Users\cheta\Desktop\cc-ass2

```
Google Cloud SDK Shell - py google-cloud-sdk\bin\dev_appserver.py C:\Users\cheta\Desktop\cc-ass2
Welcome to the Google Cloud CLI! Run "gcloud -h" to get the list of available commands.
---
C:\Program Files (x86)\Google\Cloud SDK>py google-cloud-sdk\bin\dev_appserver.py C:\Users\cheta\Desktop\cc-ass2
INFO 2022-04-25 23:54:26,285 devappserver2.py:316] Skipping SDK update check.
WARNING 2022-04-25 23:54:26,503 simple_search_stub.py:1196] Could not read search indexes from c:\users\cheta\appdata\l
ocal\temp\appengine.None\search_indexes
INFO 2022-04-25 23:54:26,503 <string>:384] Starting API server at: http://localhost:63138
INFO 2022-04-25 23:54:26,519 dispatcher.py:281] Starting module "default" running at: http://localhost:8080
INFO 2022-04-25 23:54:26,519 admin_server.py:150] Starting admin server at: http://localhost:8000
INFO 2022-04-25 23:54:30,184 instance.py:294] Instance PID: 11380
```

Step 6 – Once you have written the command in the gcloud sdk shell and press **Enter**. After a few moments, your application will start.

Step 7 – Press **Browse** to open a browser pointing at your application which is running at <http://localhost:8000/>

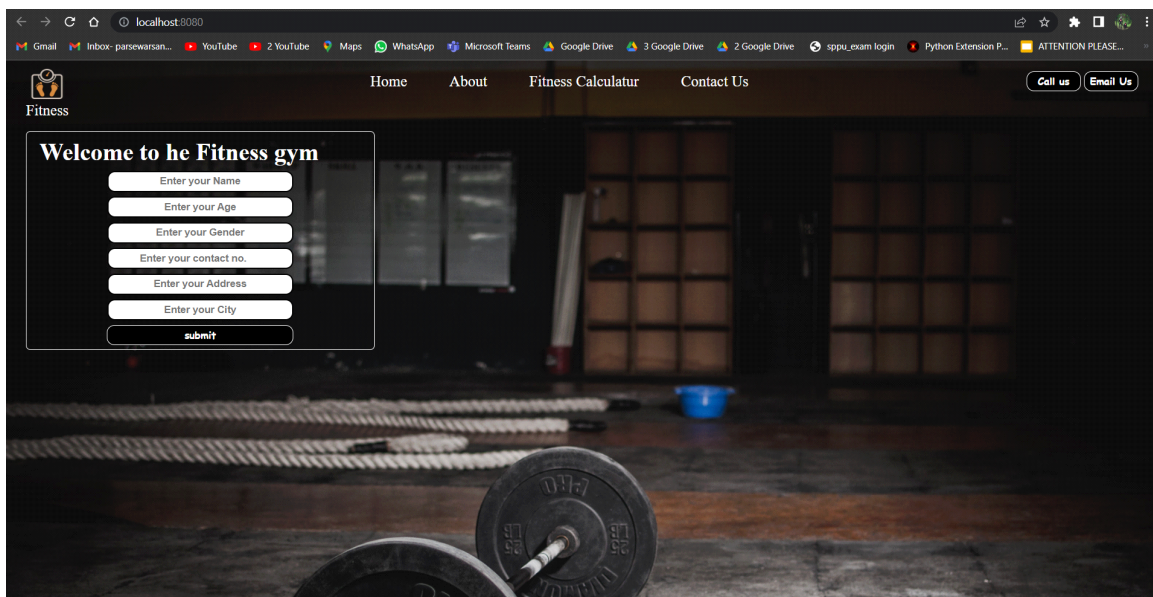
Step 8 – Now go to <http://localhost:8000> in your browser. You will see the following dashboard,



The screenshot shows the Google App Engine dashboard in a web browser. The browser's address bar shows `localhost:8000/instances`. The dashboard header includes the Google App Engine logo and "Development SDK 0.0.0". On the left, a sidebar lists various tools like "Datastore Viewer", "Interactive Console", etc. The main content area is titled "Instances" and displays a table with instance details.

	Latency (ms)	QPS	Total Requests	Runtime
default				python27
6eaddbbf7ac81c5ef49efef8b5ef63643550	0.0	0.00	0	

Step 9 – Click on the default button to run your application



Note:

- 1) When you make a mistake in the **app.yaml** file – you must fix the mistake and attempt to start the application again.
- 2) If you make a mistake in a file like **index.html**, you can simply fix the file and press refresh in your browser – there is no need to restart the server.

Conclusion:

Thus we have designed and developed a simple web application and successfully launch an application using the GAE launcher.

Assignment No. 3

Title:

Simulate a cloud scenario using CloudSim and run a scheduling algorithm that is not present in CloudSim.

Objective:

6. To learn the CloudSim tool
7. To learn how to simulate a cloud scenario using CloudSim?
8. To learn how to run a scheduling algorithm in CloudSim?

Requirements:

1. Java JDK and JRE
2. CloudSim archives (CloudSim4)
3. Eclipse IDE

Theory:**Basics of CloudSim:**

CloudSim is a simulation toolkit that supports the modeling and simulation of the core functionality of the cloud, like job/task queue, processing of events, creation of cloud entities (data-center, datacentre brokers, etc), communication between different entities, implementation of broker policies, etc. This toolkit allows to:

- Test application services in a repeatable and controllable environment.
- Tune the system bottlenecks before deploying apps in an actual cloud.
- Experiment with different workload mix and resource performance scenarios on simulated infrastructure for developing and testing adaptive application provisioning techniques

Core features of CloudSim are:

- The Support modeling and simulation of large-scale computing environment as federated cloud data centers, and virtualized server hosts, with customizable policies for provisioning host resources to virtual machines and energy-aware computational resources

- It is a self-contained platform for modeling cloud's service brokers, provisioning, and allocation policies.
- It supports the simulation of network connections among simulated system elements.
- Support for simulation of federated cloud environment, that inter-networks resources from both private and public domains.
- Availability of a virtualization engine that aids in the creation and management of multiple independent and co-hosted virtual services on a data center node.
- Flexibility to switch between the space-shared and time-shared allocation of processing cores to virtualized services.

All these features would help in accelerating the development, testing, and deployment of potential resource/application provisioning policies/ algorithms for Cloud Computing based systems.

Benefits of CloudSim:

- **No capital investment involved.** With a simulation tool like CloudSim, there is no installation or maintenance cost.
- **Easy to use and Scalable.** You can change the requirements such as adding or deleting resources by changing just a few lines of code.
- **Risks can be evaluated at an earlier stage.** In Cloud Computing utilization of real testbeds limits the experiments to the scale of the testbed and makes the reproduction of results an extremely difficult undertaking. With simulation, you can test your product against test cases and resolve issues before actual deployment without any limitations.
- **No need for try-and-error approaches.** Instead of relying on theoretical and imprecise evaluations which can lead to inefficient service performance and revenue generation, you can test your services in a repeatable and controlled environment free of cost with CloudSim.

Why use CloudSim?

- Open source and free of cost, so it favours researchers/developers working in the field.
- Easy to download and set-up.
- It is more generalized and extensible to support modelling and experimentation.
- Does not require any high-specs computer to work on.
- Provides pre-defined allocation policies and utilization models for managing resources, and allows implementation of user-defined algorithms as well.
- The documentation provides pre-coded examples for new developers to get familiar with the basic classes and functions.
- Tackle bottlenecks before deployment to reduce risk, lower costs, increase performance, and raise revenue.

CloudSim Architecture:

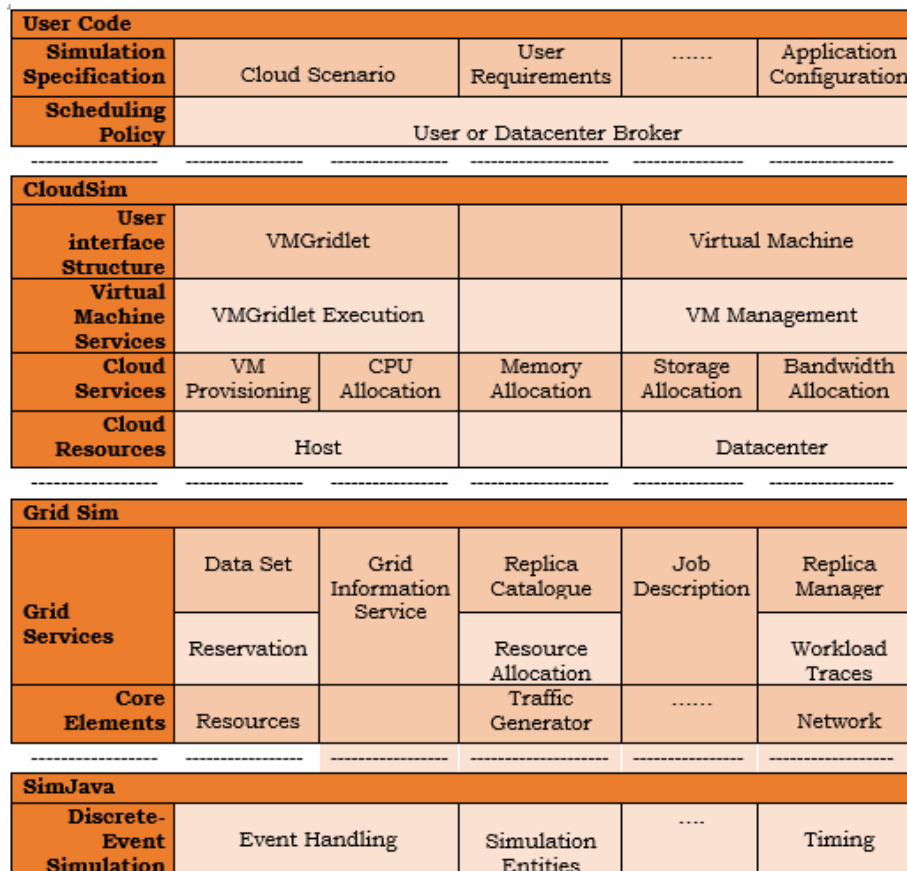


Figure 1: Layered CloudSim Architecture

Figure 1 shows the layered implementation of the CloudSim software framework and architectural components. The **CloudSim Core simulation engine** provides support for modeling and simulation of virtualized Cloud-based data center environments including queuing and processing of events, creation of cloud system entities (like data center, host, virtual machines, brokers, services, etc.) communication between components and management of the simulation clock.

At the lowest layer is the **SimJava** discrete event simulation engine that implements the core functionalities required for higher-level simulation frameworks such as queuing and processing of events, creation of system components (services, host, data center, broker, virtual machines), communication between components, and management of the simulation clock.

Next follows the libraries implementing the **GridSim** toolkit that support high-level software components for modeling multiple Grid infrastructures, including networks and associated traffic profiles, and fundamental Grid components such as the resources, data sets, workload traces, and information services.

The CloudSim is implemented at the next level by programmatically extending the core functionalities exposed by the GridSim layer. CloudSim provides novel support for the modeling and simulation of virtualized Cloud-based data centers environments such as dedicated management interfaces for VMs, memory, storage, and bandwidth.

CloudSim layer manages the instantiation and execution of core entities (VMs, hosts, data centers, applications) during the simulation period. This layer is capable of concurrently instantiating and transparently managing a large-scale Cloud infrastructure consisting of thousands of system components. The fundamental issues such as provisioning of hosts to VMs based on user requests, managing application execution, and dynamic monitoring are handled by this layer. A Cloud provider, who wants to study the efficacy of different policies in allocating its hosts, would need to implement his strategies at this layer by programmatically extending the core VM provisioning functionality. There is a clear distinction at this layer on how a host is allocated to different competing VMs in the Cloud. A Cloud host can be concurrently shared among a number of VMs that execute applications based on user-defined QoS specifications.

The top-most layer in the simulation stack is the User Code that exposes configuration-related functionalities for hosts (number of machines, their specifications, and so on), applications (number of tasks and their requirements), VMs, number of users and their application types, and broker scheduling policies. A Cloud application developer can generate a mix of user request distributions, application configurations, and Cloud availability scenarios at this layer and perform robust tests based on the custom Cloud configurations already supported within the CloudSim.

As Cloud computing is a rapidly evolving research area, there is a severe lack of defined standards, tools and methods that can efficiently tackle the infrastructure and application-level complexities. Hence in the near future, there would be a number of research efforts both in academia and industry toward defining core algorithms, policies, and application benchmarking based on execution contexts. By extending the basic functionalities already exposed by CloudSim, researchers would be able to perform tests based on specific scenarios and configurations, hence allowing the development of best practices in all the critical aspects related to Cloud Computing (*Reference: "CloudSim: A Novel Framework for Modeling and Simulation of Cloud Computing Infrastructures and Services". Rodrigo N. Calheiros, Rajiv Ranjan, César A. F. De Rose, and Rajkumar Buyya.)

Design and Implementation of CloudSim:

The fundamental classes of CloudSim Simulation Toolkit (Version 3.0.3), are displayed in the figure below, which are also the building blocks of the simulator.

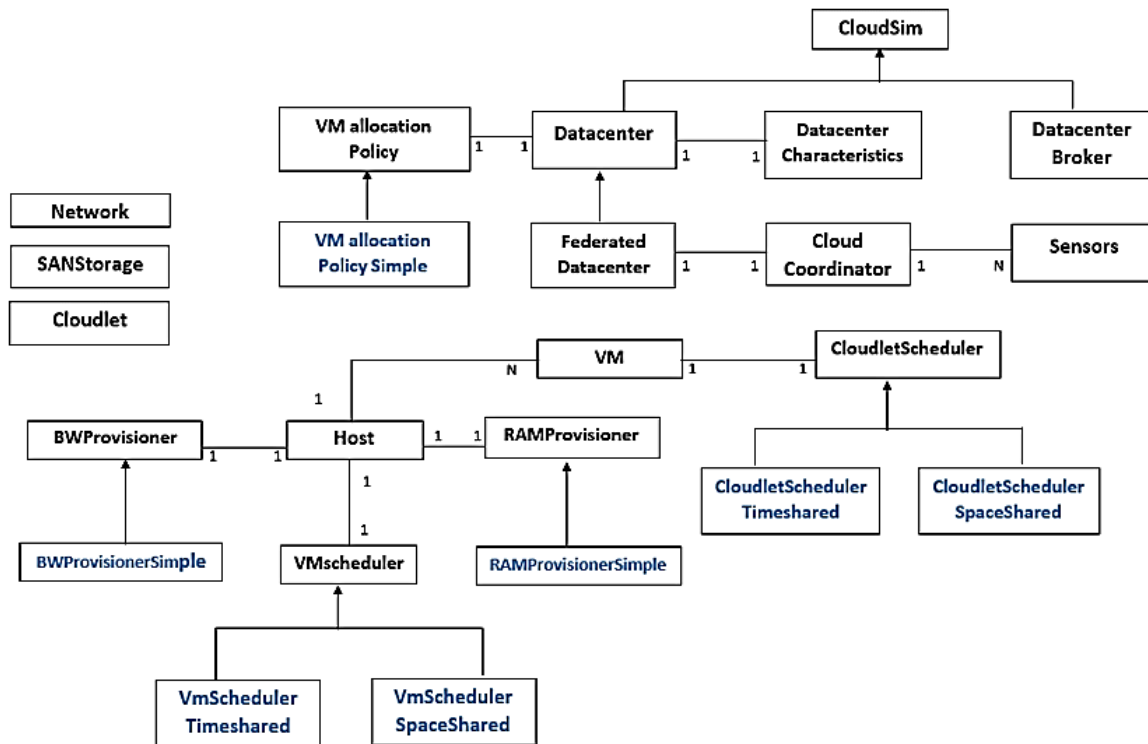


Figure 2: CloudSim class design diagram

Common classes used during simulation:

Datacenter: This class models the core infrastructure level services (hardware, software) offered by resource providers in a Cloud computing environment. This class provides methods to specify the functional requirements of the Datacenter as well as methods to set the allocation policies of the VMs etc.

Datacenter-Broker: This class models a broker, which is responsible for mediating between users and service providers depending on users' QoS requirements and deploys service tasks across Clouds. This is an entity acting on behalf of the user/customer. It is responsible for the functioning of VMs, including VM creation, management, destruction, and submission of cloudlets to the VM.

SANStorage: This class models a storage area network that is commonly available to Cloud-based data centers for storing large chunks of data. SANStorage implements a simple interface that can be used to simulate storage and retrieval of any amount of data, at any time subject to the availability of network bandwidth.

Host: This class executes actions related to the management of virtual machines. It also defines policies for provisioning memory and bandwidth to the virtual machines, as well as allocating CPU cores to the virtual machines.

Virtual Machine: this class represents a virtual machine by providing data members defining a VM's bandwidth, RAM, MIPS (million instructions per second), and size while also providing setter and getter methods for these parameters.

Cloudlet: This class models the Cloud-based application services (content delivery, social networking, business workflow), which are commonly deployed in the data centers. a cloudlet class represents any task that is run on a VM, like a processing task, or a memory access task, or a file updating task etc. It stores parameters defining the characteristics of a task such as its length, size, mi (million instructions) and provides methods similarly to VM class while also providing methods that define a task's execution time, status, cost and history.

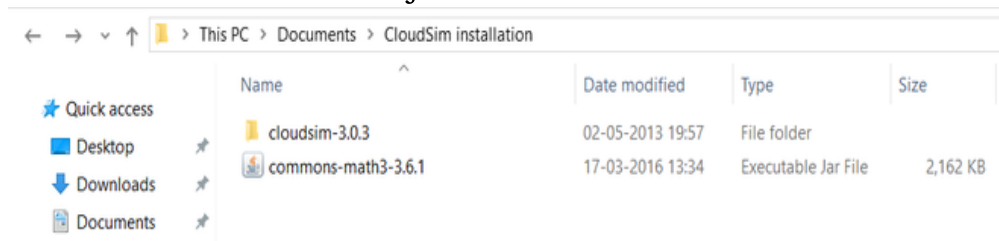
CloudSim: this is the class responsible for initializing and starting the simulation environment after all the necessary cloud entities have been defined and later stopping after all the entities have been destroyed.

(*Reference: "CloudSim: A Novel Framework for Modeling and Simulation of Cloud Computing Infrastructures and Services". Rodrigo N. Calheiros, Rajiv Ranjan, César A. F. De Rose, and Rajkumar Buyya.)

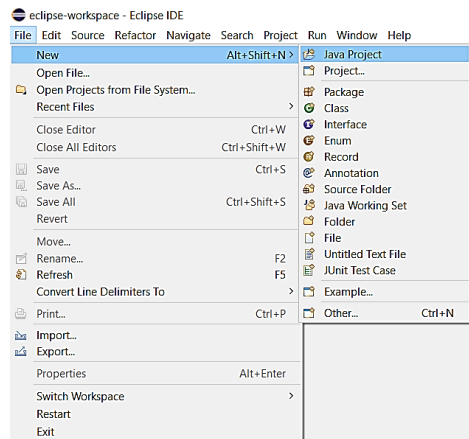
Installation of CloudSim and creation of simulation environment:

Steps :

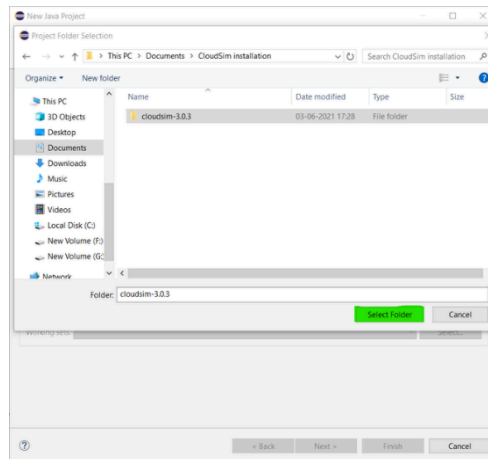
1. Visit <https://github.com/Cloudslab/cloudsim/releases> to download the CloudSim archives for CloudSim 3.0.3 and above.
2. Extracts cloudsim-3.0.3 into a folder. Also, extract the commons-math3-3.6.1 jar into the same folder.



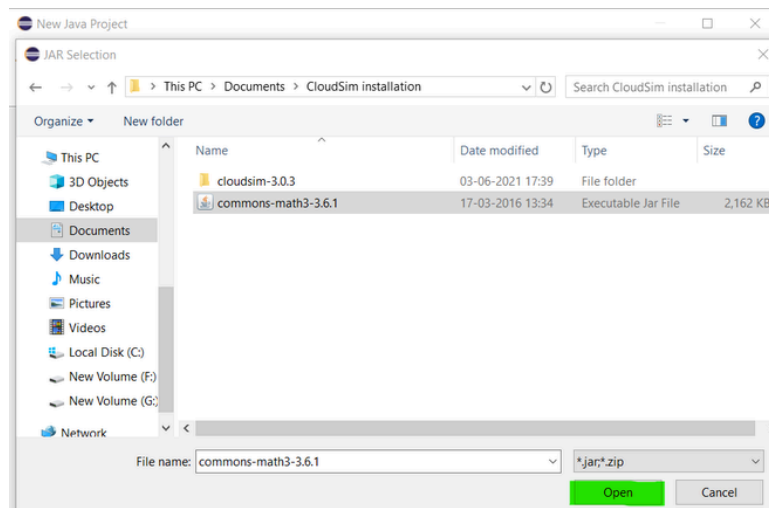
3. The jars folder of the extracted archive should contain the following files:
 - a) cloudsim-4.0.jar
 - b) cloudsim-examples.jar
4. Create a new Project: Open Eclipse IDE and go to File -> New -> Java Project.



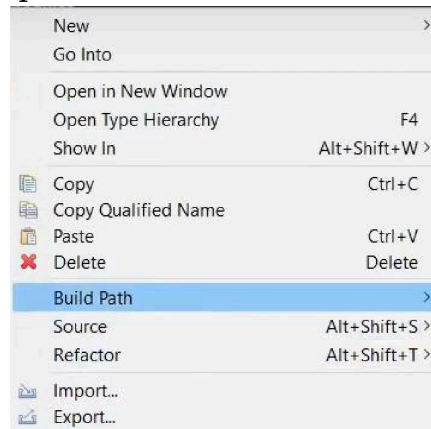
5. Name your project and then uncheck the Use default location box just under it and click on Browse. Browse to the folder where you extracted your files and select the cloudsim-3.0.3 folder.



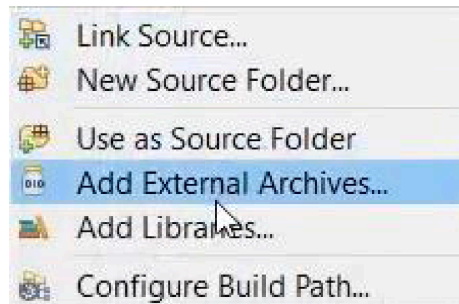
6. Click Next and go to Libraries -> Add External JARs. Now browse to the same folder where you extracted your commons-math3 jar file and Open it.



7. Right click on the project root and select the Build Path option from the dropdown.



8. Select the Configure Build Path section from the extended dropdown



9. Create a new package in the src directory of the project.
10. Copy the code files for the constants, Data Center Creator, Data Center Broker, Matrix Generator and the SJF scheduler files from the <https://github.com/suyash-more/Cloud-Computing-Projects/tree/master/Scheduling-Algorithm-in-CloudSim/src>
11. Make sure that the package name provided in each file is the same as the previously created package.

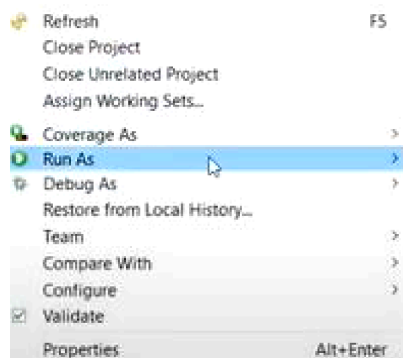
SJF algorithm:

1. SJF stands for Shortest Job First
2. Shortest Job first has the advantage of having a minimum average waiting time among all scheduling algorithms.
3. It is a Greedy Algorithm.
4. It may cause starvation if shorter processes keep coming. This problem can be solved using the concept of ageing.
5. It is practically infeasible as Operating System may not know burst time and therefore may not sort them. While it is not possible to predict execution time, several methods can be used to estimate the execution time for a job, such as a weighted average of previous execution times. SJF can

be used in specialized environments where accurate estimates of running time are available.

Execution of code:

Right-click on the project



Run the project as a Java Application (option in the extended dropdown)



The result is displayed on the console

```
Starting SJF Scheduler...
Initializing new Matrices...
Initialising...
Starting CloudSim version 3.0
Datacenter_0 is starting...
Datacenter_1 is starting...
Datacenter_2 is starting...
Datacenter_3 is starting...
Datacenter_4 is starting...
Broker_0 is starting...
Entities started.
0.0: Broker_0: Cloud Resource List received with 5 resource(s)
0.0: Broker_0: Trying to Create VM #2 in Datacenter_0
0.0: Broker_0: Trying to Create VM #3 in Datacenter_1
0.0: Broker_0: Trying to Create VM #4 in Datacenter_2
0.0: Broker_0: Trying to Create VM #5 in Datacenter_3
0.0: Broker_0: Trying to Create VM #6 in Datacenter_4
0.1: Broker_0: VM #2 has been created in Datacenter #2, Host #0
0.1: Broker_0: VM #3 has been created in Datacenter #3, Host #0
0.1: Broker_0: VM #4 has been created in Datacenter #4, Host #0
0.1: Broker_0: VM #5 has been created in Datacenter #5, Host #0
0.1: Broker_0: VM #6 has been created in Datacenter #6, Host #0
0.1: Broker_0: Sending cloudlet 0 to VM #5
0.1: Broker_0: Sending cloudlet 1 to VM #6
0.1: Broker_0: Sending cloudlet 2 to VM #4
0.1: Broker_0: Sending cloudlet 3 to VM #3
0.1: Broker_0: Sending cloudlet 4 to VM #5
0.1: Broker_0: Sending cloudlet 5 to VM #2
0.1: Broker_0: Sending cloudlet 6 to VM #6
0.1: Broker_0: Sending cloudlet 7 to VM #3
0.1: Broker_0: Sending cloudlet 8 to VM #4
0.1: Broker_0: Sending cloudlet 9 to VM #4
0.1: Broker_0: Sending cloudlet 10 to VM #3
0.1: Broker_0: Sending cloudlet 11 to VM #2
0.1: Broker_0: Sending cloudlet 12 to VM #6
0.1: Broker_0: Sending cloudlet 13 to VM #4
```

```
1292.724: Broker_0: Cloudlet 0 received
1907.232: Broker_0: Cloudlet 5 received
2260.772: Broker_0: Cloudlet 1 received
2784.16: Broker_0: Cloudlet 3 received
2903.4: Broker_0: Cloudlet 2 received
3065.932: Broker_0: Cloudlet 11 received
4113.036: Broker_0: Cloudlet 4 received
4485.576: Broker_0: Cloudlet 18 received
4837.776: Broker_0: Cloudlet 20 received
4956.164: Broker_0: Cloudlet 6 received
5643.272: Broker_0: Cloudlet 8 received
5807.608: Broker_0: Cloudlet 21 received
6354.656: Broker_0: Cloudlet 7 received
6905.915999999999: Broker_0: Cloudlet 23 received
7719.535999999999: Broker_0: Cloudlet 24 received
8614.368: Broker_0: Cloudlet 12 received
8752.444: Broker_0: Cloudlet 10 received
8986.24: Broker_0: Cloudlet 9 received
10703.216: Broker_0: Cloudlet 15 received
10857.967999999999: Broker_0: Cloudlet 14 received
11948.996: Broker_0: Cloudlet 25 received
13310.556: Broker_0: Cloudlet 13 received
13635.776: Broker_0: Cloudlet 16 received
15582.328: Broker_0: Cloudlet 17 received
16230.772: Broker_0: Cloudlet 19 received
17007.956: Broker_0: Cloudlet 27 received
19003.152000000002: Broker_0: Cloudlet 22 received
19533.66: Broker_0: Cloudlet 28 received
22878.644: Broker_0: Cloudlet 26 received
25918.7: Broker_0: Cloudlet 29 received
25918.7: Broker_0: All Cloudlets executed. Finishing...
```

===== OUTPUT =====							
Cloudlet ID	STATUS	Data center ID	VM ID	Time	Start Time	Finish Time	Waiting Time
00	SUCCESS	05	05	1292.62	00.1	1292.72	00
05	SUCCESS	02	02	1907.13	00.1	1907.23	00
01	SUCCESS	06	06	2260.67	00.1	2260.77	00
03	SUCCESS	03	03	2784.06	00.1	2784.16	00
02	SUCCESS	04	04	2903.3	00.1	2903.4	00
11	SUCCESS	02	02	1158.7	1907.23	3065.93	1907.13
04	SUCCESS	05	05	2820.31	1292.72	4113.04	1292.62
18	SUCCESS	02	02	1419.64	3065.93	4485.58	3065.83
20	SUCCESS	02	02	352.2	4485.58	4837.78	4485.48
06	SUCCESS	06	06	2695.39	2260.77	4956.16	2260.67
08	SUCCESS	04	04	2739.87	2903.4	5643.27	2903.3
21	SUCCESS	05	05	1694.57	4113.04	5807.61	4112.94
07	SUCCESS	03	03	3570.5	2784.16	6354.66	2784.06
23	SUCCESS	02	02	2068.14	4837.78	6905.92	4837.68
24	SUCCESS	02	02	813.62	6905.92	7719.54	6905.82
12	SUCCESS	06	06	3658.2	4956.16	8614.37	4956.06
10	SUCCESS	03	03	2397.79	6354.66	8752.44	6354.56
09	SUCCESS	04	04	3342.97	5643.27	8986.24	5643.17
15	SUCCESS	06	06	2088.85	8614.37	10703.22	8614.27
14	SUCCESS	03	03	2105.52	8752.44	10857.97	8752.34
25	SUCCESS	03	03	1091.03	10857.97	11949	10857.87
13	SUCCESS	04	04	4324.32	8986.24	13310.56	8986.14
16	SUCCESS	06	06	2932.56	10703.22	13635.78	10703.12
17	SUCCESS	06	06	1946.55	13635.78	15582.33	13635.68
19	SUCCESS	04	04	2920.22	13310.56	16230.77	13310.46
27	SUCCESS	06	06	1425.63	15582.33	17007.96	15582.23
22	SUCCESS	04	04	2772.38	16230.77	19003.15	16230.67
28	SUCCESS	06	06	2525.7	17007.96	19533.66	17007.86
26	SUCCESS	04	04	3875.49	19003.15	22878.64	19003.05
29	SUCCESS	04	04	3040.06	22878.64	25918.7	22878.54

Makespan using SJF: 4396.012266367984

Conclusion: Thus we have successfully execute the scheduling algorithm using cloudSim.

Assignment No. 4

Title:

Find a procedure to transfer the files from one virtual machine to another virtual machine.

Objective:

1. To learn about Virtual machines (VM).
2. To learn the Basics of the File transfer mechanism.
3. To transfer the files from one VM to another VM.

Requirements:

1. Two Virtual machines installed. (For example, we have Ubuntu 21 and Kali Linux for this assignment)

Theory:

What is a virtual machine?

A virtual machine is an application providing a platform-independent programming runtime that allows applications to execute in the same manner on different platforms. The virtual machine acts as a bridge to the real environment, hiding the details of the operating system. Do not confuse this term with system virtual machines, such as VMware, Virtual server, Xen which enables one to run multiple OS on a single piece of hardware.

Types of virtualization techniques:

1. Guest Operating system virtualization
2. Shared Kernel Virtualization
3. Kernel Level Virtualization
4. Hypervisor Virtualization

Types of virtual machines:

We can classify virtual machines into two types:

System Virtual Machine:

These types of virtual machines give us complete system platform and gives the execution of the complete virtual operating system. Just like virtual box, system virtual machine is providing an environment for an OS to be installed completely. We can see in below image that our hardware of Real Machine is being distributed between two simulated operating systems by Virtual machine monitor. And then some programs, processes are going on in that distributed hardware of simulated machines separately.

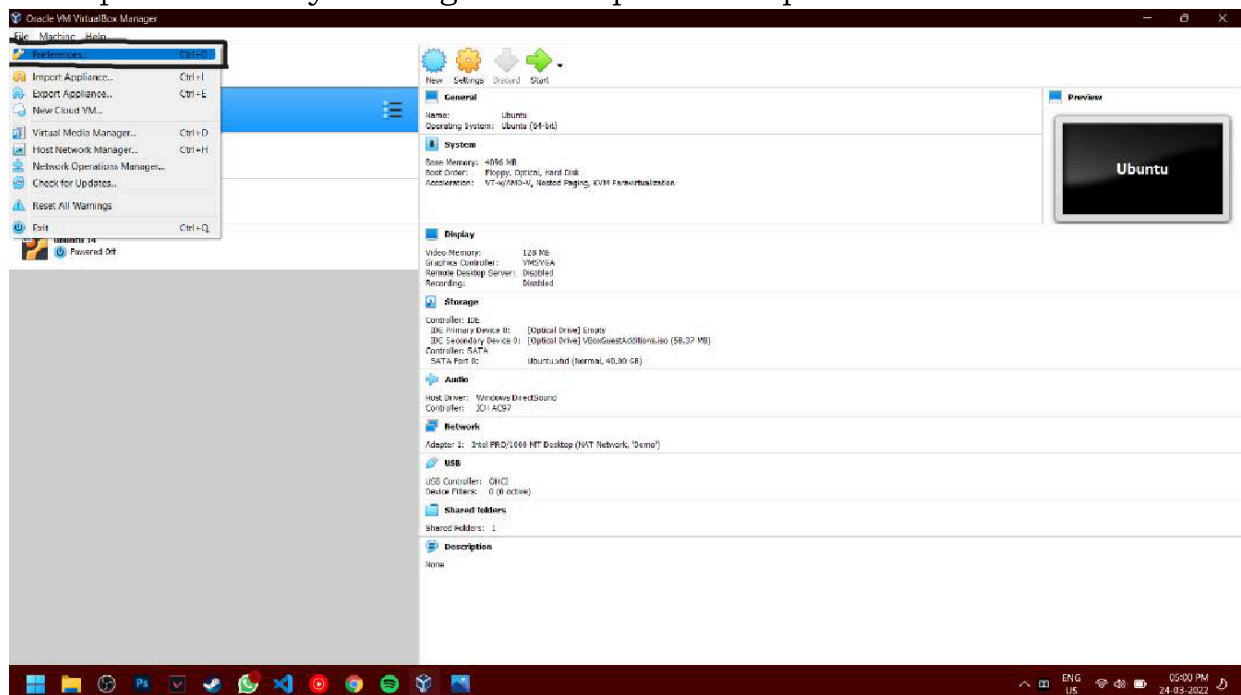
Process Virtual Machine:

While process virtual machines, unlike system virtual machines, do not provide us with the facility to install the virtual operating system completely. Rather it creates a virtual environment of that OS while using some app or program and this environment will be destroyed as soon as we exit from that app. Like in the below image, there are some apps running on the main OS as well some virtual machines are created to run other apps. This shows that as those programs required different OS, process virtual machine provided them with that for the time being those programs are running.

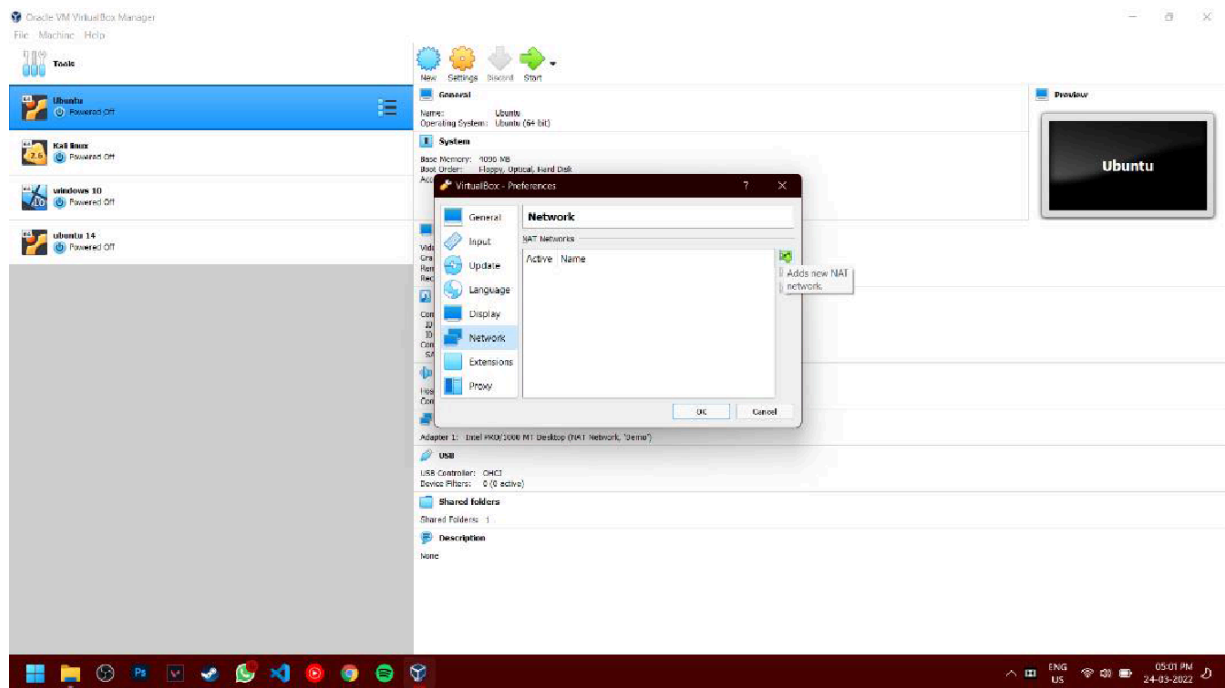
Steps:

Step 1. Create a Nat network in which 2 virtual machines can communicate.

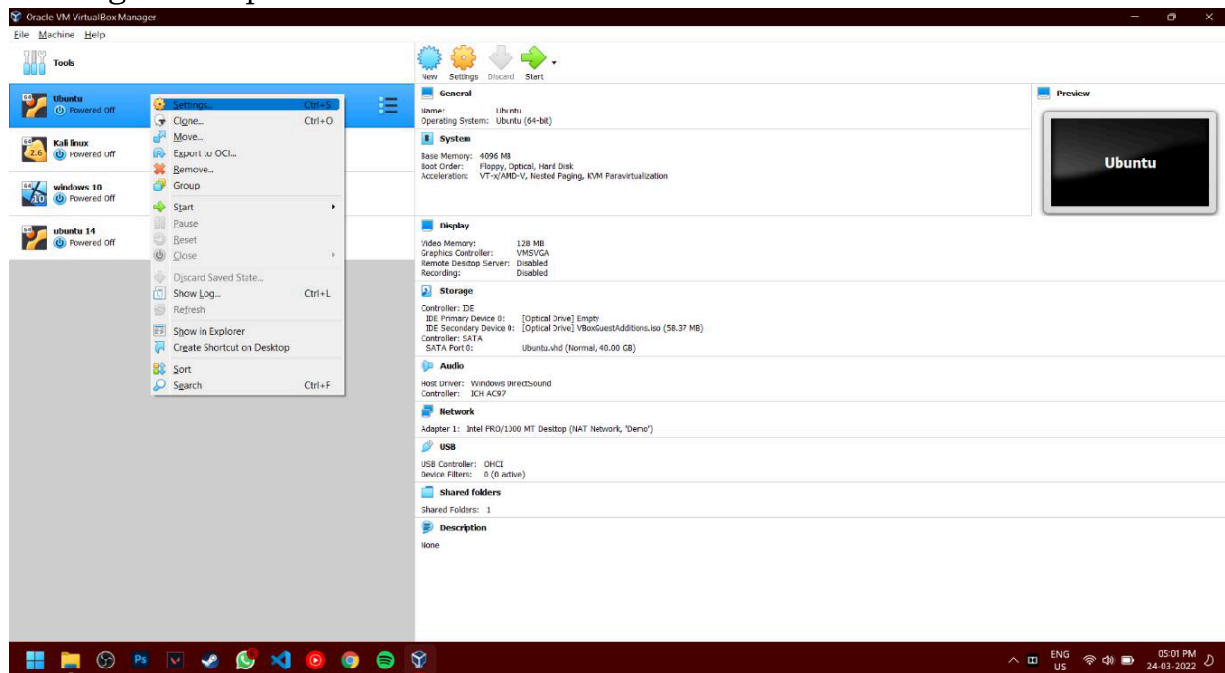
Go to preferences by clicking the File option in Top.



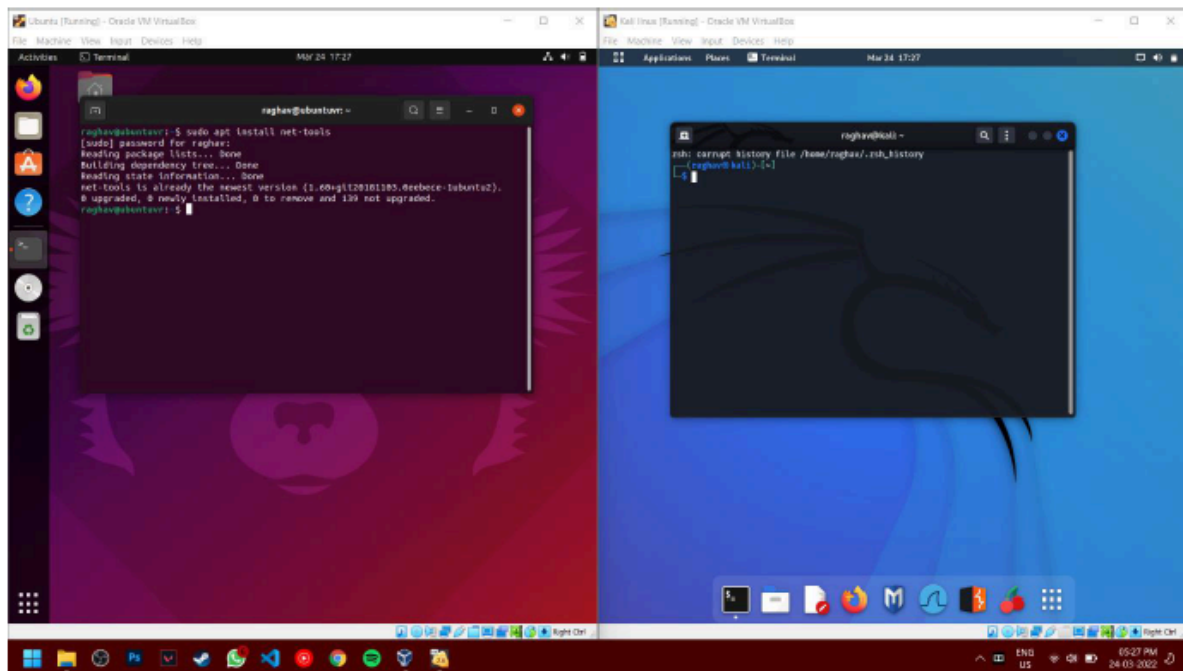
Now select the network option and create a new NAT network



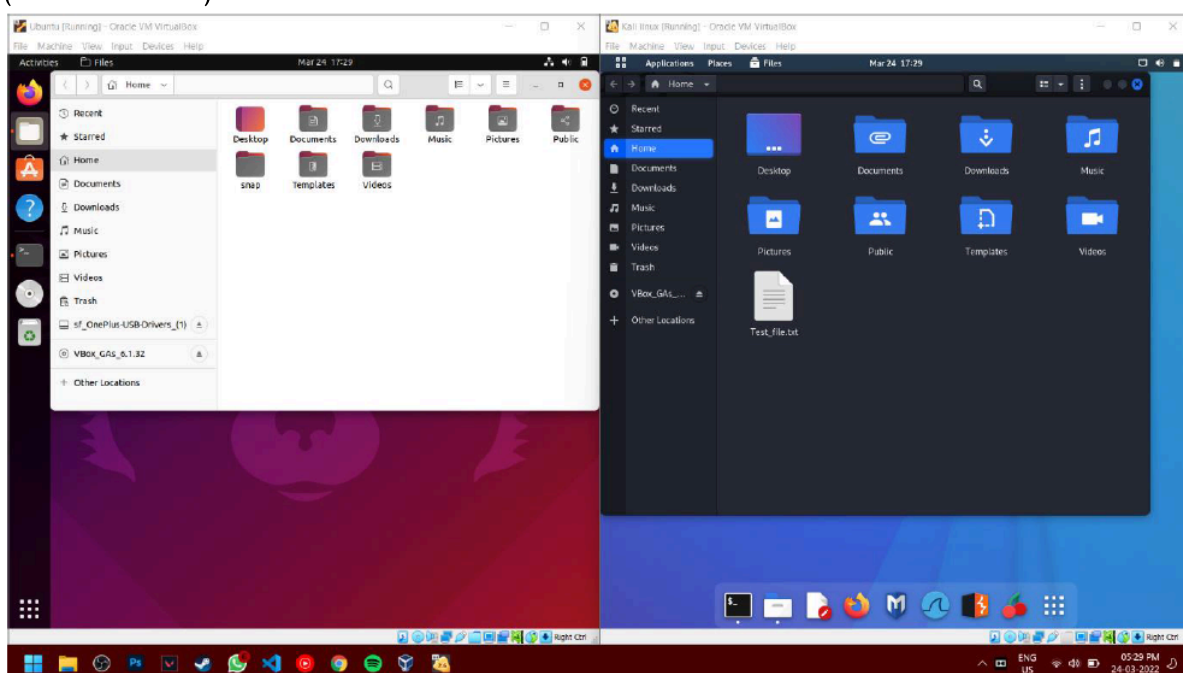
After creating a NAT network, now go to virtual machine setting by right clicking on the preferred machine.



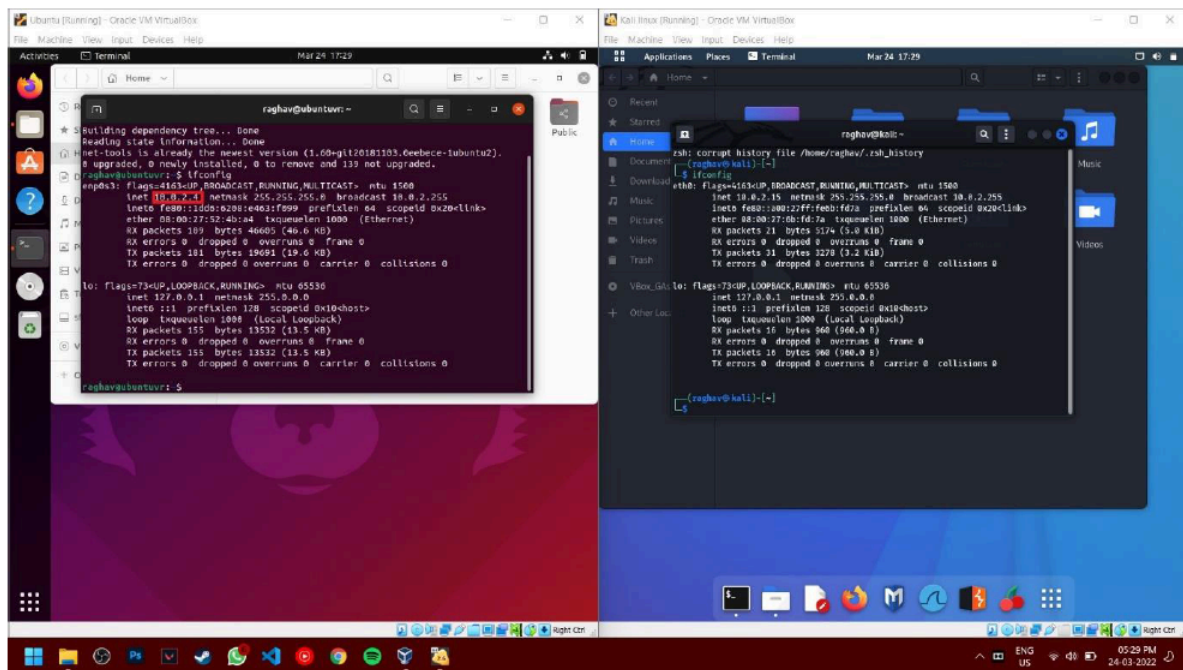
Now go to Network and change the attached option to “NAT network” and select the network we created earlier.



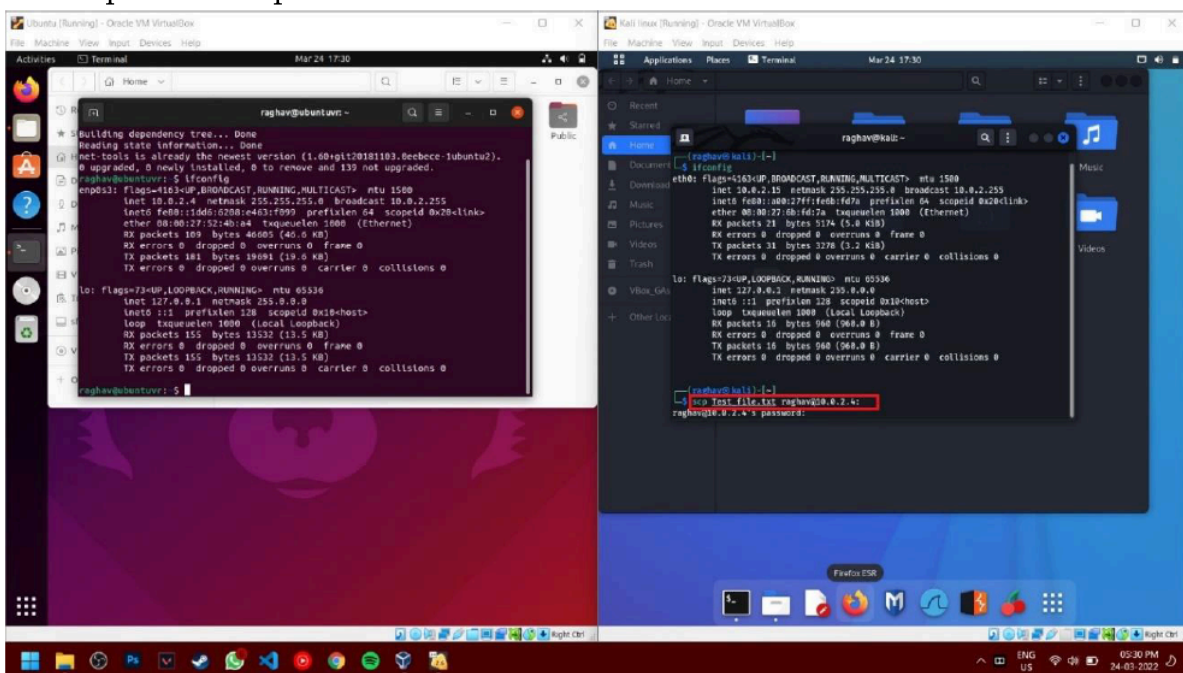
Now create a file in Home folder using any text editor. Here we have used “Test_file.txt” and will transfer from Kali Linux (Right Machine) to Ubuntu (left Machine).



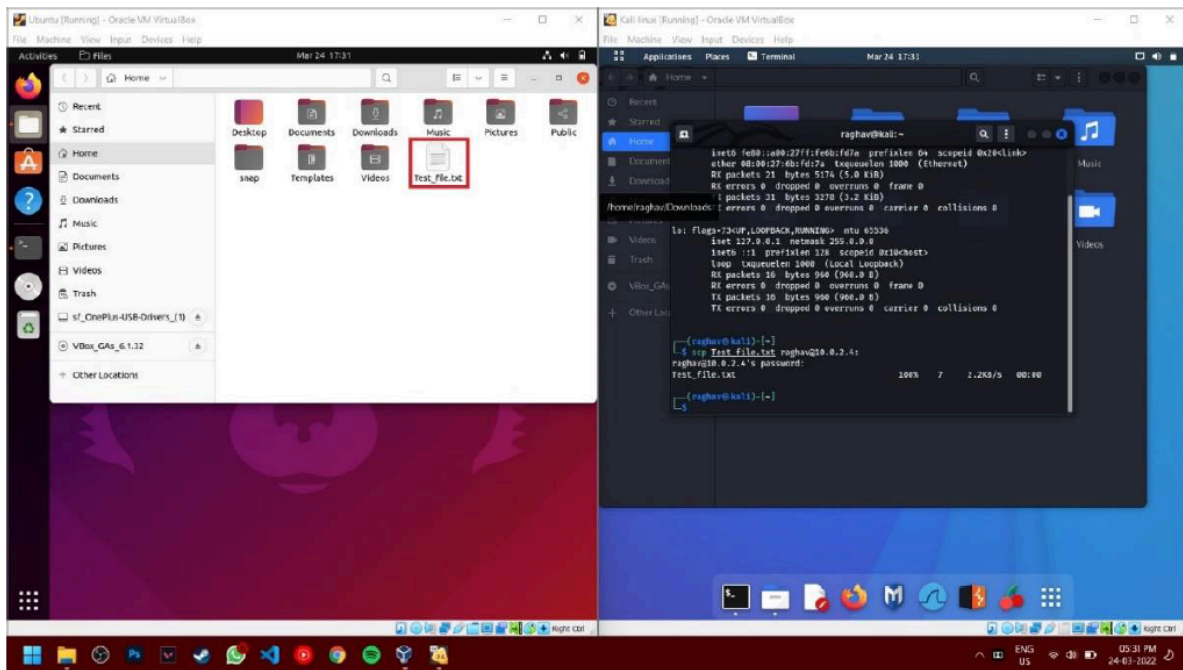
Now we can check i/p address of Ubuntu where we want to transfer the file using “ifconfig” command. Here Ubuntu has i/p address 10.0.2.4.



Transfer the file using command- `scp Test_file.txt raghav@10.0.2.4:`
 Where Test_file.txt is our file
 raghav is the username of the Ubuntu
 10.0.2.4 is ip address of Ubuntu (left machine)
 Optional: if scp is not installed then install by using the command:
`sudo apt install openssh-server`

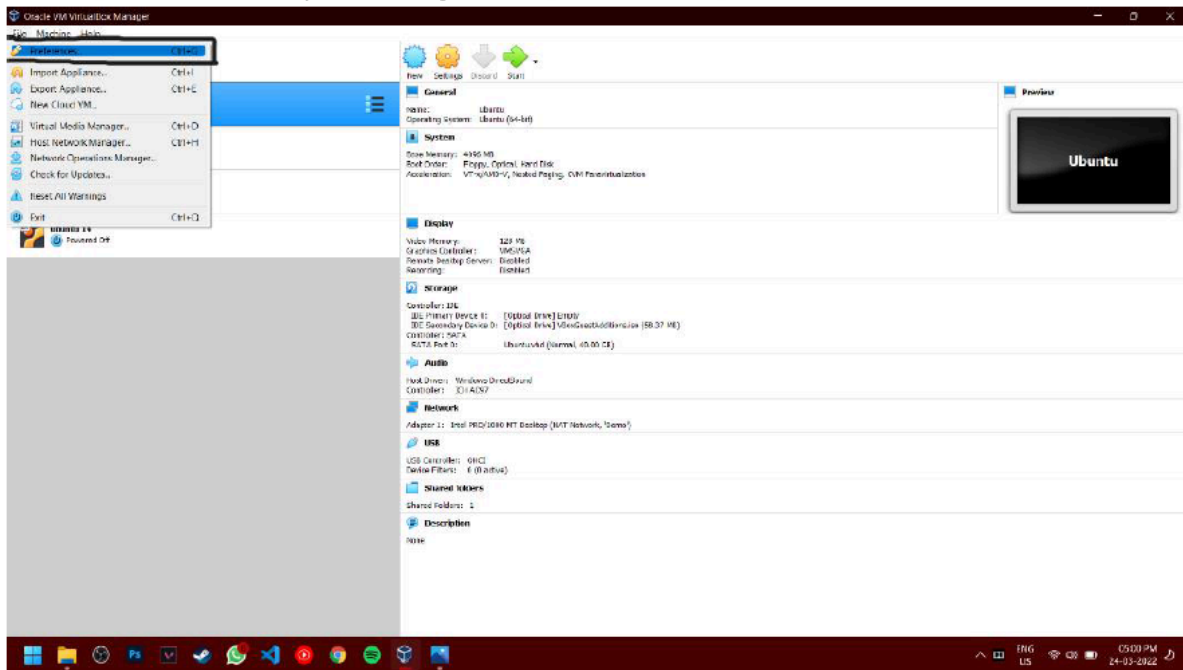


Now enter the password for Ubuntu (left machine) admin, after enter the password, the file will be sent from Kali Linux (Right machine) to Ubuntu (Left machine)

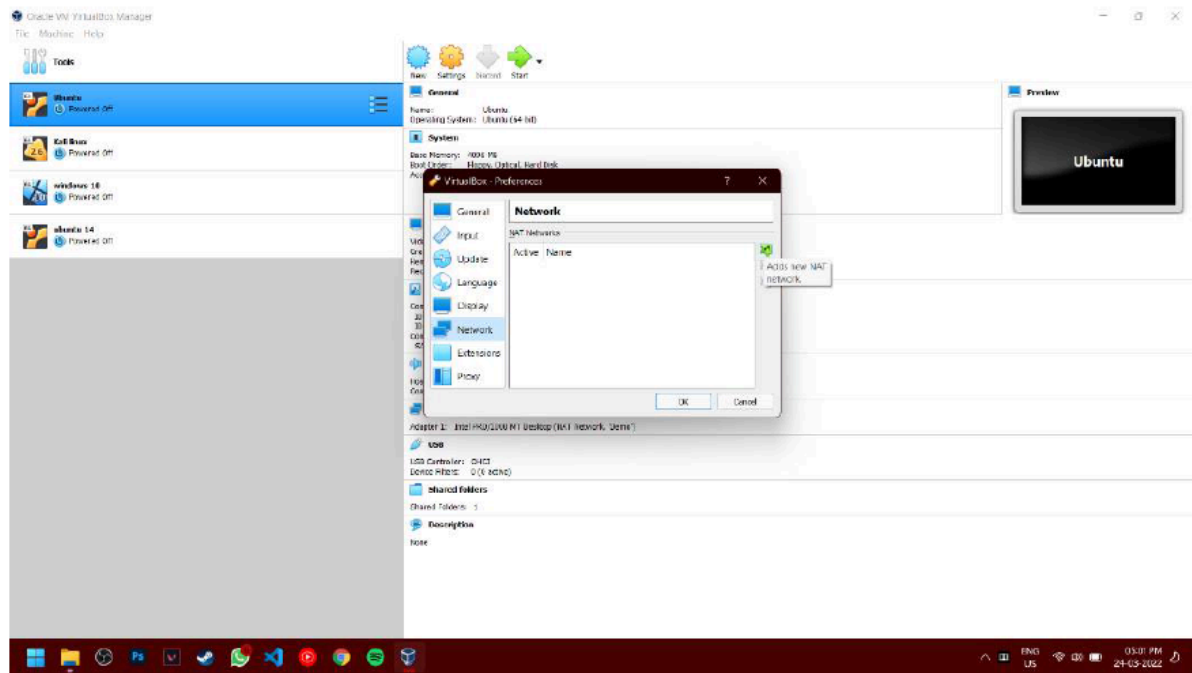


Step 3: Create a NAT network in which 2 virtual machine can communicate.

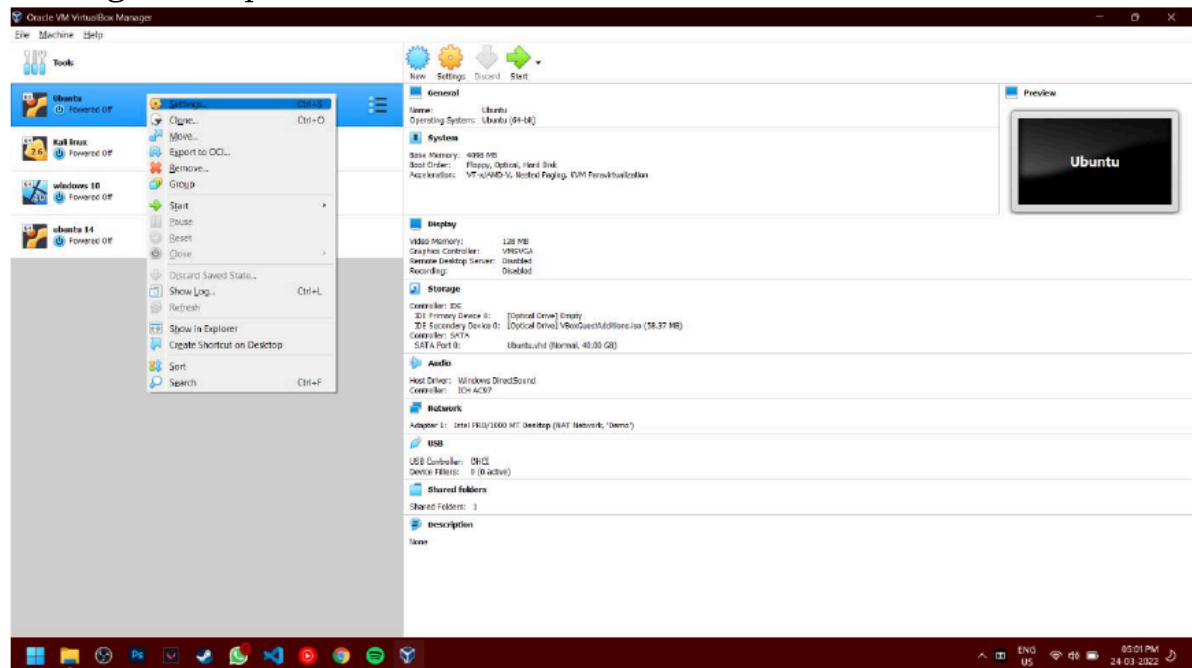
Go to preferences by clicking File option in Top.



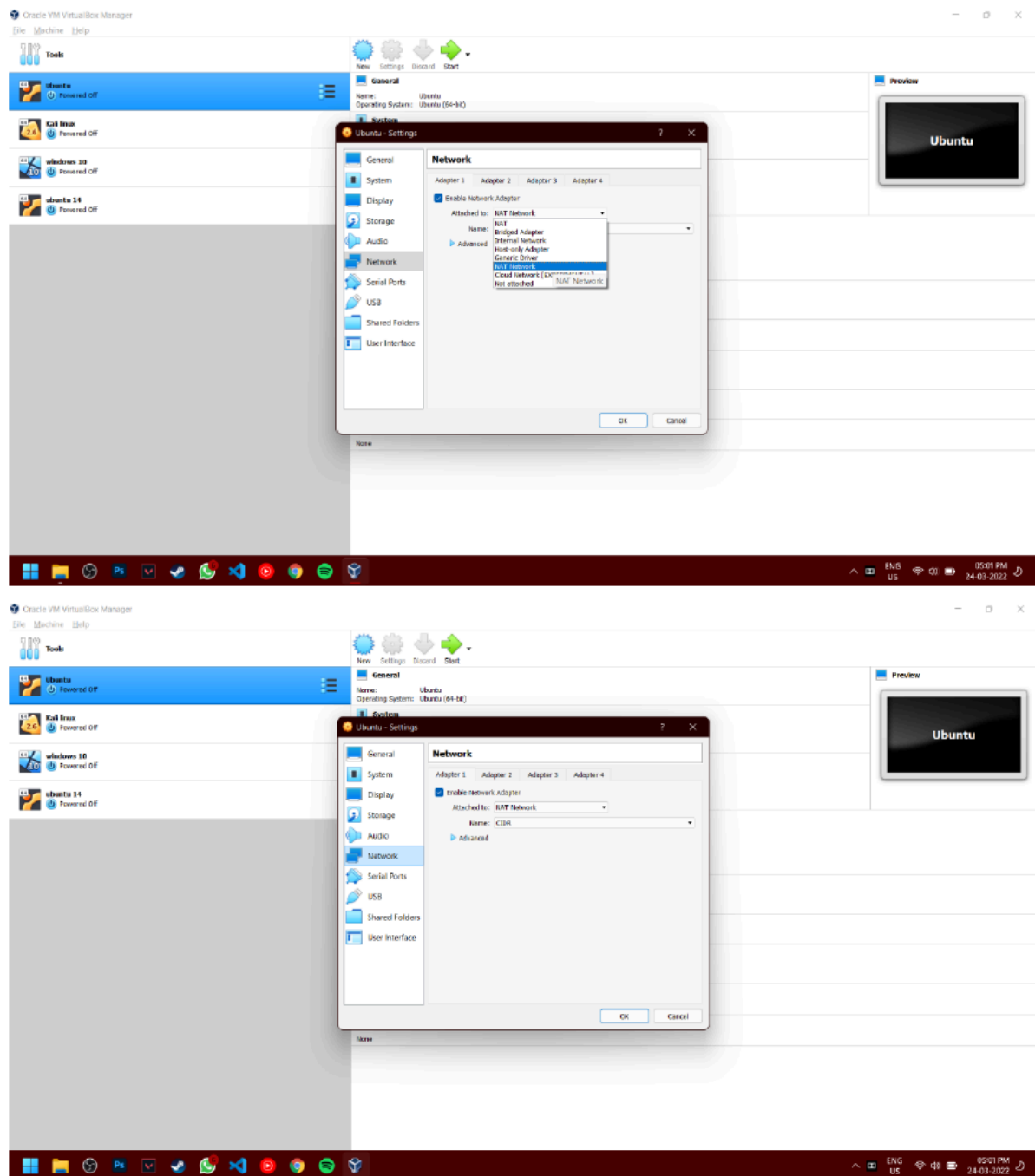
Now select the network option and create a new NAT network



After creating a NAT network, now go to virtual machine setting by right clicking on the preferred machine.



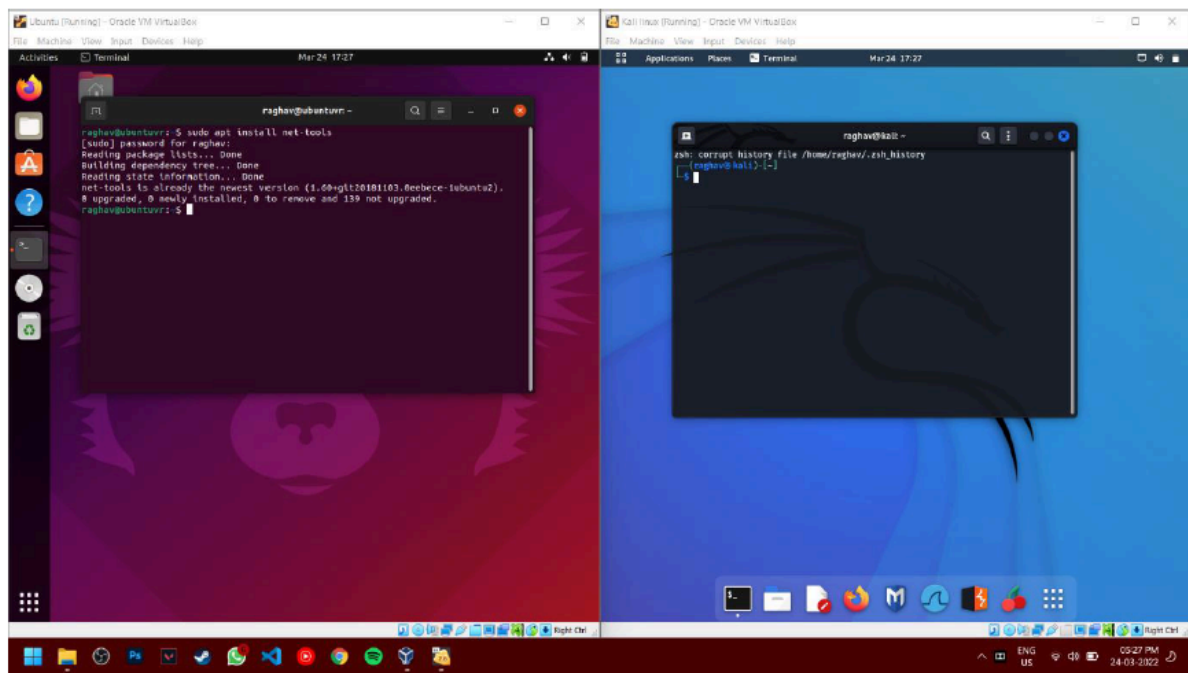
Now go to Network and change the attached option to “NAT network” and select the network we created earlier.



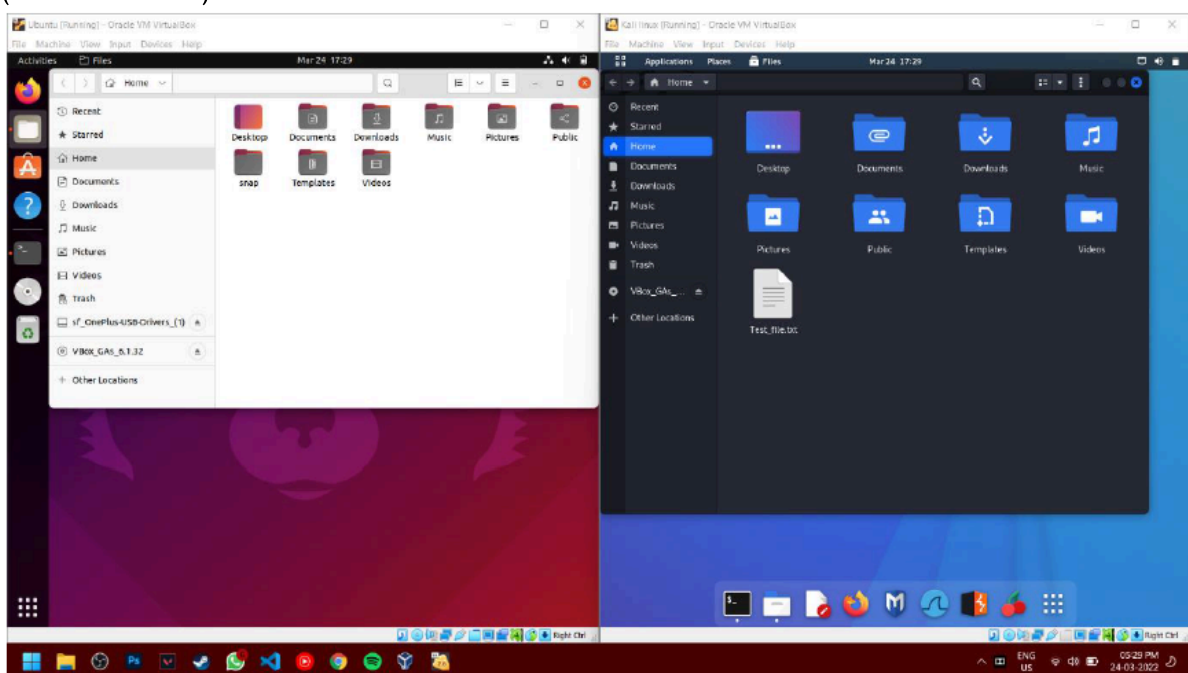
Now repeat the same process for another machine.

Step 4: Launch both virtual machines

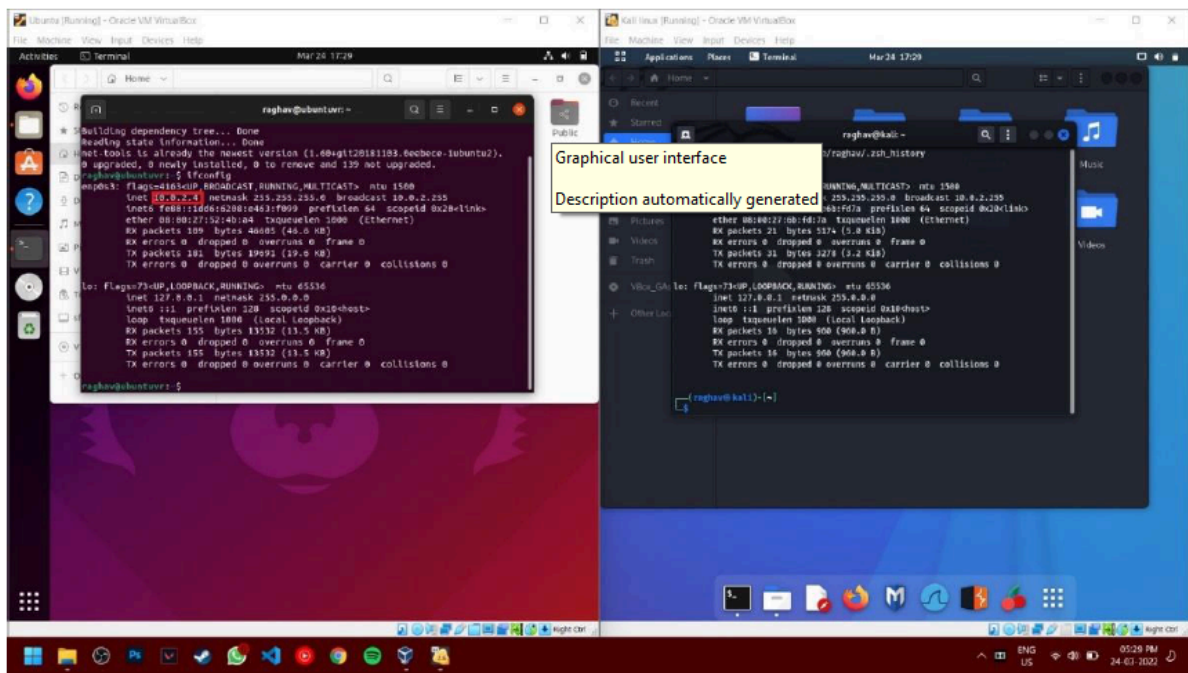
Now install “Net-tools” on both machine which will help to identify i/p address of the machine. Command: `sudo apt install net-tools`



Now create a file in Home folder using any text editor. Here we have used “Test_file.txt” and will transfer from Kali Linux (Right Machine) to Ubuntu (left Machine).



Now we can check i/p address of Ubuntu where we want to transfer the file using “ifconfig” command. Here Ubuntu has i/p address 10.0.2.4.



Transfer the file using command- scp Test_file.txt raghav@10.0.2.4:

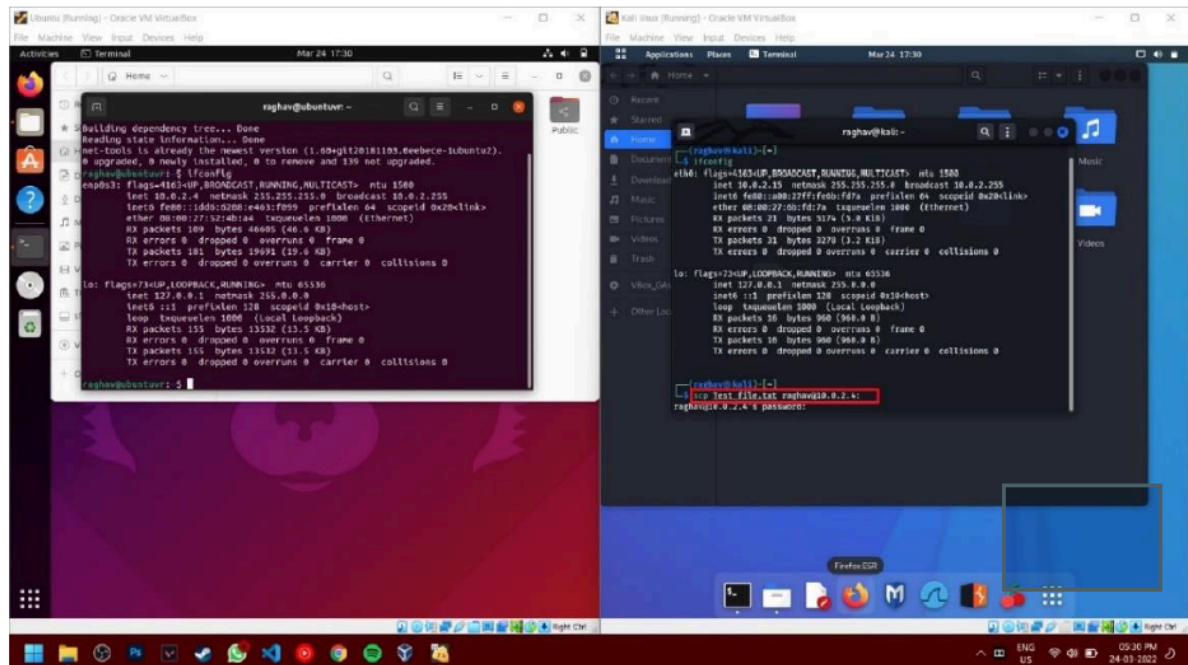
Where Test_file.txt is our file

raghav is username of Ubuntu

10.0.2.4 is ip address of Ubuntu (left machine)

Optional: if scp is not installed then install by using command:

sudo apt install openssh-server



Conclusion: Thus we have successfully transferred a file from one VM to another VM.

Assignment No. 5

Title :

Find a procedure to launch a virtual machine using trystack
(Online Openstack Demo Version)

Objective:

1. To learn Virtual Machine's working procedure.
2. To learn the Launching procedure of VM.
3. To Learn OpenStack and trystack technology

Theory:**What is a Virtual Machine?**

A virtual Machine is like a fake computer system operating on your hardware. It partially uses the hardware of your system (like CPU, RAM, disk space, etc.) but its space is completely separated from your main system. Two virtual machines don't interrupt each other's working and functioning nor they can excess each other's space which gives an illusion that we are using a totally different hardware system.

Virtual machines allow you to run an operating system in an app window on your desktop that behaves like a full, separate computer. You can use them to play around with different operating systems, run software your main operating system can't, and try out apps in a safe, sandboxed environment.

A virtual machine app creates a virtualized environment—called, simply enough, a virtual machine—that behaves like a separate computer system, complete with virtual hardware devices. The VM runs as a process in your current operating system window. You can boot an operating system installer disc (or live CD) inside the virtual machine, and the operating system will be “tricked” into thinking it's running on a real computer. It will install and run just as it would on a real, physical machine. Whenever you want to use the operating system, you can open the virtual machine program and use it in a window on your current desktop.

In the VM world, the operating system actually running on your computer is called the host, and any operating systems running inside VMs are called guests. The main purpose of VMs is to operate multiple operating systems at the same time, from the same piece of hardware.

Types of Virtual Machines

You can classify virtual machines into two types:

1. System Virtual Machine

These types of virtual machines give us a complete system platform and give the execution of the complete virtual operating system. Just like a virtual box,

A system virtual machine is providing an environment for an OS to be installed completely. The image below shows how Real Machine is being distributed between two simulated operating systems by a Virtual machine monitor. And then some programs, and processes are going on in that distributed hardware of simulated machines separately.

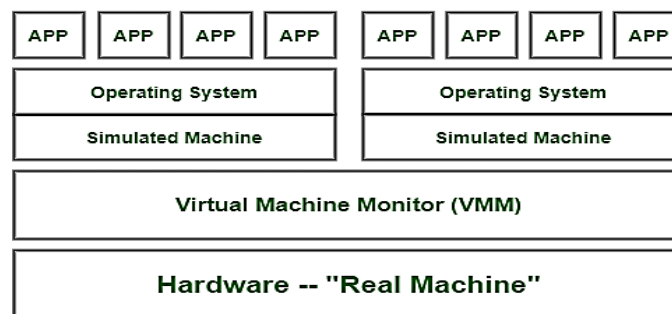


Figure 1: System Virtual machine

2. Process Virtual Machine

While process virtual machines, unlike system virtual machines, do not provide us with the facility to install the virtual operating system completely.

Rather it creates a virtual environment of that OS while using some app or program and this environment will be destroyed as soon as we exit from that app.

Like in the image below, some apps are running on the main OS and some virtual machines are created to run other apps. This shows that as those programs required different OS, process virtual machine provided them with that for the time being those programs are running.

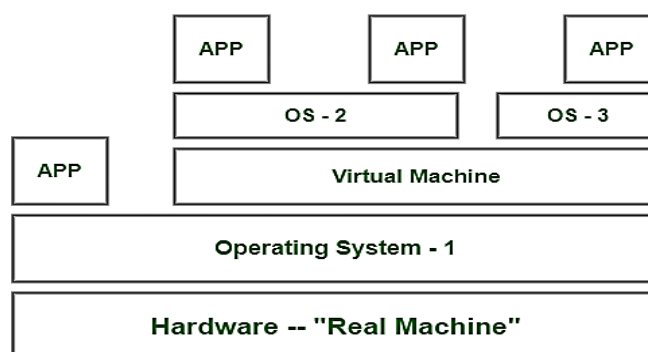


Figure 2: Process Virtual Machine

How does Virtual Machine work?

Virtualisation Technology allows you to share a system with many virtual environments. The hypervisor manages the hardware and separates the

physical resources from the virtual environments. Resources are partitioned as needed from the physical environment to the VMs.

When the VM is running and a user or program issues an instruction that requires additional resources from the physical environment, the hypervisor schedules the request to the physical system's resources so that the virtual machine's operating system and applications can access the shared pool of physical resources.

Software called hypervisors separates the physical resources from the virtual environments—the things that need those resources. Hypervisors can sit on top of an operating system (like on a laptop) or be installed directly onto hardware (like a server), which is how most enterprises virtualize. Hypervisors take your physical resources and divide them up so that virtual environments can use them.

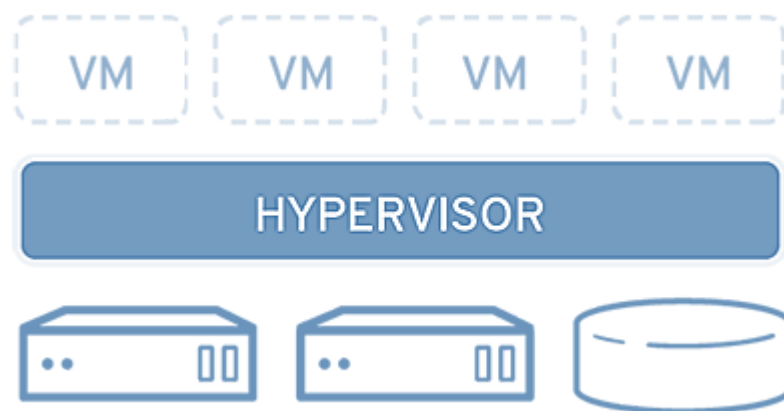


Figure 3: Hypervisor in VM

Resources are partitioned as needed from the physical environment to the many virtual environments. Users interact with and run computations within the virtual environment (typically called a guest machine or virtual machine). The virtual machine functions as a single data file. And like any digital file, it can be moved from one computer to another, opened in either one, and be expected to work the same.

The hypervisor relays the request to the physical system and caches the changes—which all happen at close to native speed (particularly if the request is sent through an open-source hypervisor based on KVM, the Kernel-based Virtual Machine).

Advantages of virtual machines

Virtual machines are great for splitting physical resources to be used by smaller virtual computers. Instead of having computers with largely unutilized resources, it is possible to distribute the CPU and RAM across multiple virtual computers. When done properly, this can significantly reduce the cost of infrastructure.

Virtual machines can also be part of server provision programs. Because virtual machines are stored on a hard drive, it is easy to copy them to recreate identical environments over and over.

Finally, the isolation provided by virtualization can let system administrators run riskier operations in a virtual environment that can easily be thrown away.

What is Open Stack?

OpenStack is a cloud operating system that controls large pools of computing, storage, and networking resources throughout a data center, all managed and provisioned through APIs with common authentication mechanisms.

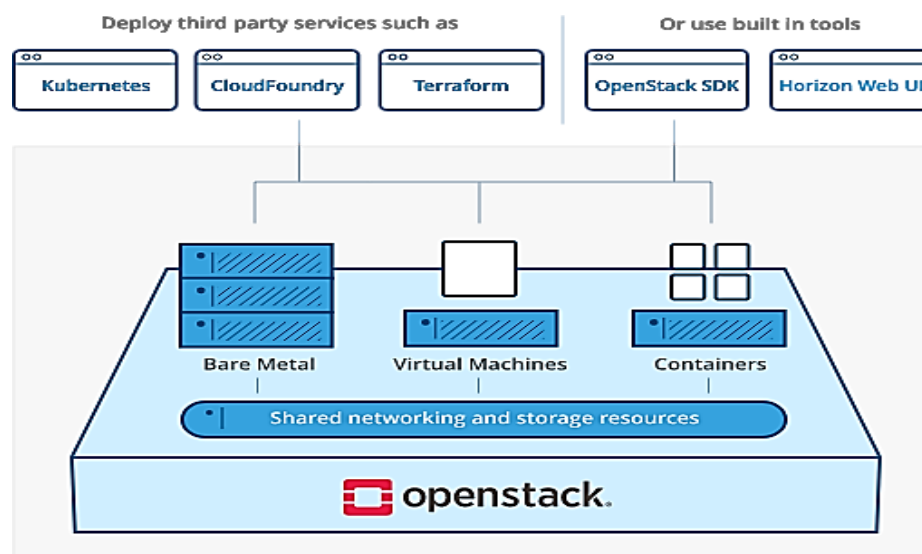


Figure 4: Open Stack

Vision for OpenStack Clouds:

- Purpose: This is a living document. Its purpose is to document the OpenStack community's vision for the output of the OpenStack project as a whole, as it evolves over time.
- Scope: The scope of this document is limited to the cloud services that an end-user interacts with. This corresponds to the main 'OpenStack' bucket and parts of the 'OpenStack Operations' bucket in the OpenStack project map.
- The Pillars of Cloud: There are at least as many different opinions of what 'cloud' means as there are software developers. However, we can all agree that cloud does mean something. Cloud computing promotes more efficient utilization of resources by reducing the transaction costs involved in provisioning and deprovisioning infrastructure to near zero, and it is able to do so because it differs in

qualitative ways from previous models of computing (including virtualization).

- Self-service: Clouds are self-service. They provide users with the ability to deploy applications on demand without having to wait for human action or review in the loop. The cloud has no ticket trackers.

What is Trystack?

TryStack is a free and easy way for users to try out OpenStack, and set up their own cloud with networking, storage, and computer instances. TryStack is an OpenStack Foundation project; they oversee the political side of it. There are different vendors and companies that have donated resources, such as rack and power and network and servers.

TryStack users will be able to launch OpenStack Compute instances that last up to 24 hours, at which point resources for the instances are reclaimed and made available to other TryStack users. Each registered user will receive a set amount of Stack Dollars they can use to “lease” instances within that time period.

TryStack is intended to provide users the ability to launch instances in one of several TryStack zones, representing different OpenStack reference architectures and geographical locations. The first zone available now has 156 cores, 1040GB memory, and 59.1 TB of disk storage running the latest OpenStack release (code-name Diablo), Dell™ PowerEdge C6100 and C6105 servers, and libvirt/KVM. Individual contributors from Dell, HP Cloud Services, NTT, and Rackspace Cloud Builders helped deploy, test, and administer TryStack for community use.

Steps to launch Virtual Machine using Trystack:

Pre-Requisites:

- 1) An account on Account on AWS or Azure or Google Cloud
- 2) OpenStack Facebook group membership.

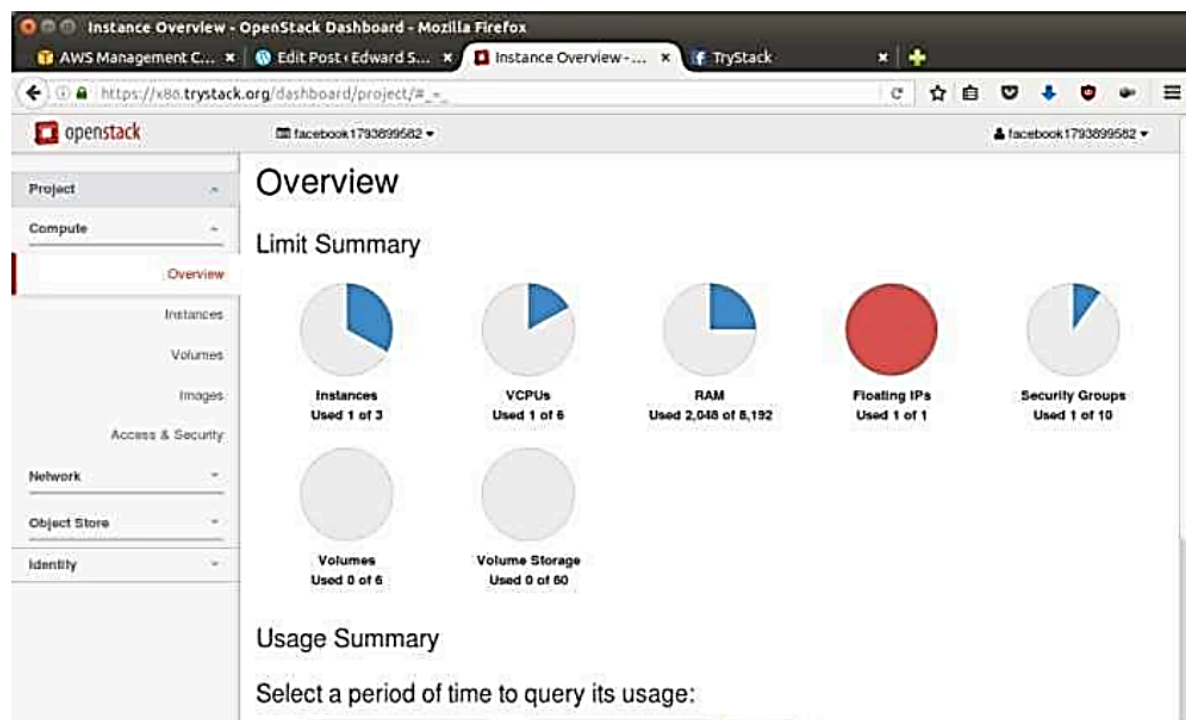
OpenStack is an open-source software cloud computing platform. OpenStack is primarily used for deploying an infrastructure as a service (IaaS) solution like Amazon Web Service (AWS). In other words, you can make your own AWS by using OpenStack. If you want to try out OpenStack, TryStack is the easiest and free way to do it.

In order to try OpenStack in TryStack, you must register yourself by joining TryStack Facebook Group. The acceptance of group needs a couple days because it's approved manually. After you have been accepted in the TryStack Group, you can log in TryStack.

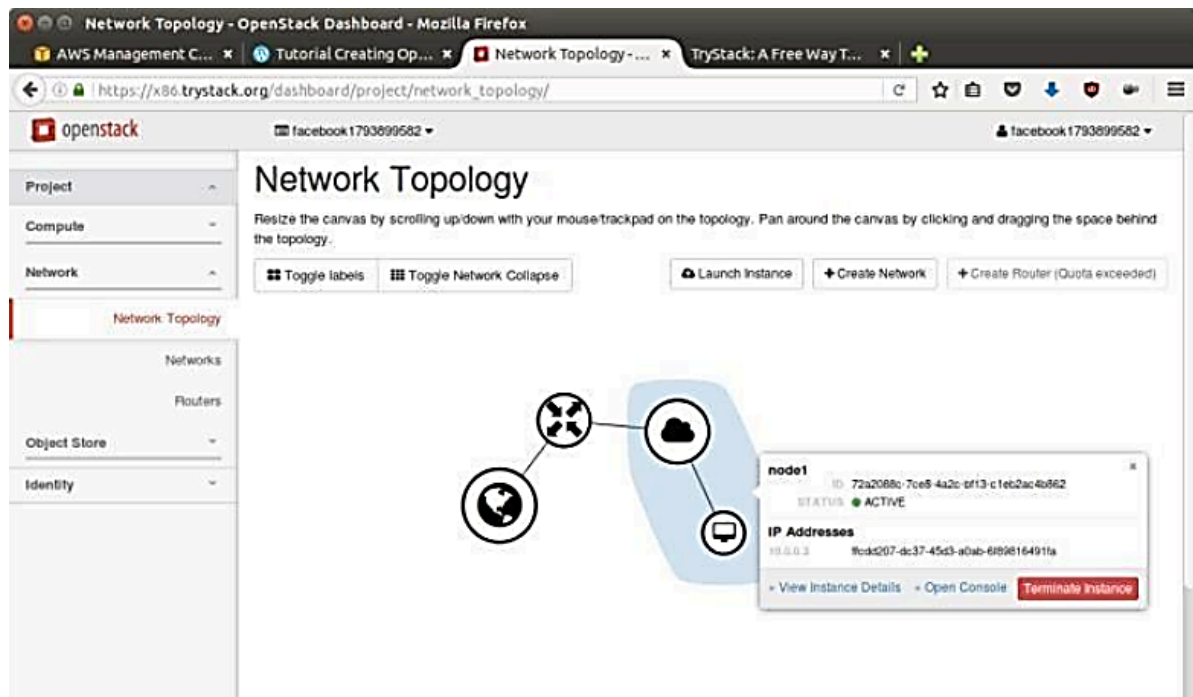
Join the Facebook Group and login to the dashboard (TryStack.org Homepage)



After you log in to the TryStack, you will see the Compute Dashboard like: (OpenStack Compute Dashboard)



To run an OpenStack instance. The instance will be accessible through the internet (have a public IP address). The final topology will like



As shown, the instance will be connected to a local network and the local network will be connected to the Internet.

Step 1: Create a Network

Network? Yes, the network here is our own local network. So, your instances will be not mixed up with the others. You can imagine this as your own LAN (Local Area Network) in the cloud.

1. Go to **Network > Networks** and then click **Create Network**.
2. In the **Network** tab, fill **Network Name** for example internal, and then click **Next**.
3. In the Subnet tab,
 1. Fill the **Network Address** with the appropriate CIDR, for example, 192.168.1.0/24. Use private network CIDR block as the best practice.
 2. Select the **IP Version** with the appropriate IP version, in this case, IPv4.
 3. Click **Next**.
4. In the **Subnet Details** tab, fill **DNS Name Servers** with 8.8.8.8 (Google DNS) and then click **Create**.

Step 2: Create Instance

Now, we will create an instance. The instance is a virtual machine in the cloud, like AWS EC2. You need the instance to connect to the network that we just created in the previous step.

1. Go to **Compute > Instances** and then click **Launch Instance**.

2. In **Details** tab,
 1. Fill **Instance Name**, for example Ubuntu 1.
 2. Select **Flavor**, for example, m1.medium.
 3. Fill **Instance Count** with **1**.
 4. Select **Instance Boot Source** with **Boot from Image**.
 5. Select **Image Name** with **Ubuntu 14.04 amd64 (243.7 MB)** if you want, install Ubuntu 14.04 in your virtual machine.
3. In **Access & Security** tab,
 1. Click **[+]** button of **Key Pair** to import the key pair. This key pair is a public and private key that we will use to connect to the instance from our machine.
2. In the **Import Key Pair** dialog,
 1. Fill the **Key Pair Name** with your machine name (for example Edward-Key).
 2. Fill **Public Key** with your **SSH public key** (usually is in `~/.ssh/id_rsa.pub`). See the description in the Import Key Pair dialog box for more information. If you are using Windows, you can use **Puttygen** to generate key pairs.
 3. Click **Import key pair**.
3. In **Security Groups**, mark/check **default**.
4. In the **Networking** tab, 1. In **Selected Networks**, select networks that have been created in Step 1, for example internal.
5. Click **Launch**.
6. If you want to create multiple instances, you can repeat steps 1-5. I created one more instance with the instance name Ubuntu 2.

Step 3: Create a Router

I guess you already know what a router is. In step 1, we created our network, but it is isolated. It doesn't connect to the Internet. To make our network has an internet connection, we need a router that runs as the gateway to the internet.

1. Go to **Network > Routers** and then click **Create Router**.
2. Fill **Router Name** for example router1 and then click **Create router**.
3. Click on your **router name link**, for example router1, **Router Details** page.
4. Click **Set Gateway** button in the upper right:
 1. Select **External networks** with **external**.
 2. Then OK.
5. Click **Add Interface** button.
 1. Select **Subnet** with the network that you have been created in Step 1.
 2. Click **Add interface**.
6. Go to **Network > Network Topology**. You will see the network topology. In the example, there are two network, i.e. external and internal, those are bridged by a router. There are instances those are joined to internal network.

Step 4: Configure Floating IP Address

Floating IP address is public IP address. It makes your instance is accessible from the internet. When you launch your instance, the instance will have a private network IP, but no public IP. In OpenStack, the public

Ips is collected in a pool and managed by admin (in our case is TryStack). You need to request a public (floating) IP address to be assigned to your instance.

1. Go to **Compute > Instance**.
2. In one of your instances, click **More > Associate Floating IP**.
3. In **IP Address**, click Plus [+].
4. Select **Pool** to **external** and then click **Allocate IP**.
5. Click **Associate**.
6. Now you will get a public IP, e.g. 8.21.28.120, for your instance.

Step 5: Configure Access & Security

OpenStack has a feature like a firewall. It can whitelist/blacklist your in/out connection. It is called *Security Group*.

1. Go to **Compute > Access & Security** and then open **Security Groups** tab.
2. In **default** row, click **Manage Rules**.
3. Click **Add Rule**, choose **ALL ICMP** rule to enable ping into your instance, and then click **Add**.
4. Click **Add Rule**, choose **HTTP** rule to open HTTP port (port 80), and then click **Add**.
5. Click **Add Rule**, choose **SSH** rule to open SSH port (port 22), and then click **Add**.
6. You can open other ports by creating new rules.

Step 6: SSH to Your Instance

Now, you can SSH your instances to the floating IP address that you got in the step 4. If you are using Ubuntu image, the SSH user will be ubuntu.

Conclusion:

We have launch a virtual machine using Openstack trystack.

Assignment No. 6

Title:

Design and deploy a web application in a PaaS environment.

Objective:

1. To understand the concept of Platform as a service.
2. To understand how we can deploy any web-application on PaaS.

Prerequisites:

1. A basic understanding of web application development
2. Access to a PaaS platform (such as Heroku, Azure App Service, or Google App Engine)

Theory:**PAAS (Platform as a service):**

Platform as a Service (PaaS) refers to a cloud computing configuration that helps enterprises operate with an efficient cloud-based strategy. PaaS provides a platform for customers to develop, run, and manage applications without building and maintaining the cloud infrastructure required to develop and launch applications. PaaS permits more efficient application development since the organization can focus on the application itself.

Working of PAAS:

Platform-as-a-service (PaaS) is a type of cloud computing model in which a service provider delivers a platform to customers. The platform enables the organization to develop, run, and manage business applications without the need to build and maintain the infrastructure such software development processes require.

PaaS is offered via a service provider's hosted cloud infrastructure. Users typically access PaaS offerings via a web browser. Customers pay for PaaS on a per-use basis. Some providers will charge a flat monthly fee for access to the platform and applications hosted on the platform.

PaaS can be delivered through public, private, or hybrid clouds. With a public cloud PaaS, the customer controls software deployment while the cloud provider delivers all the major IT components needed for running applications. These components can include servers, storage systems, networks, operating systems, and databases. With a private cloud offering, PaaS is delivered as software or an appliance behind a customer's firewall, typically in its on-premises data center. Hybrid cloud PaaS offers a mix of the two types of cloud service.

Rather than replace an organization's entire IT infrastructure for software development, PaaS provides key services such as application hosting or Java development. Some PaaS offerings include application design,

development, testing, and deployment. PaaS services can also include web service integration, development team collaboration, database integration, and information security. PaaS includes multiple underlying cloud infrastructure components, including servers, networking equipment, operating systems, storage services, middleware, and databases.

All of these technology offerings are owned, operated, configured, and maintained by the service providers. Customers can avoid having to lay out investments in these foundational IT components that they might not be able to use to the fullest extent possible. PaaS also includes widely used resources such as development tools, programming languages, libraries, containers, database management systems, and other tools.

Advantages of PAAS:

Platform as a Service hardware and software provide benefits like streamlining development tools, reducing infrastructure cost, working on multiple operating systems, and supporting various programming languages.

Platform as a Service benefits outlined by research firm Gartner include:

- API development and management – Companies can use PaaS solutions to manage application programming interfaces as well as microservices. This includes security, development, creating new APIs, and end-to-end API management.
- Business analytics/intelligence – Some PaaS solutions include tools that empower enterprises to analyze their data for business insights and patterns of behavior. These tools give the organization the required information to make better decisions and more accurately predict things like market demand for products.
- Business process management (BPM) – Organizations can access a BPM platform delivered as a service through PaaS solutions. BPM suites integrate IT components needed for process management, including data, business rules, and service-level agreements.
- Communications – PaaS can deliver communications platforms. This allows developers to add communication features to applications, such as voice, video, and messaging.
- Databases – A PaaS provider can deliver database services to an organization, such as set-up and maintenance. Database PaaS is an on-demand, secure, and scalable self-service database model. Provisioning can be automated, as well as administration of databases, according to analyst firm Forrester Research.
- Primary data management (PDM) – Primary data management (MDM/PDM) software tracks the most essential company-wide data points, providing a single point of reference for data. From this point of reference, the software provides insights related to company operations, clients, and goals. Such data might include reference data such as information about customer transactions, and analytical data to support decision making.

Users can then implement that data as they see fit, keep records of data history, and make projections based on findings.

Examples of PAAS:

Following are some examples of PASS:

1. Google App Engine:
2. Amazon Amplify:
3. Digital Ocean:
4. Dokku:
5. North-Flank:
6. Heroku:
7. AWS Elastic Beanstalk:

Steps:

Step 1: Design the Web Application

1. Choose a programming language and framework for your web application. Some popular options include Ruby on Rails, Node.js with Express, or Python with Django.
2. Identify the key features and functionalities of your web application.
3. Create a wireframe or mockup of your web application to visualize its layout and design.

Step 2: Develop the Web Application

1. Write the code for your web application using your chosen programming language and framework.
2. Test your web application locally to ensure that it is functioning properly.

Step 3: Deploy the Web Application to a PaaS Platform

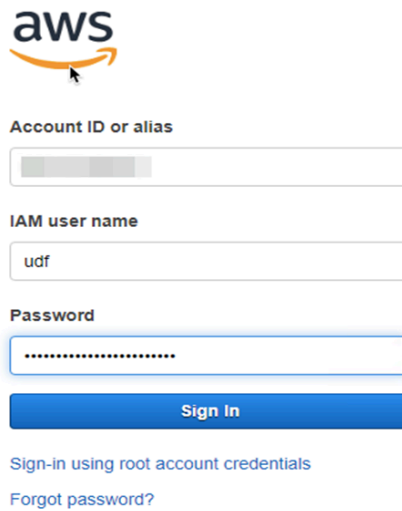
1. Create an account on a PaaS platform of your choice.
2. Create a new application on the PaaS platform and configure its settings.
3. Upload your web application code to the PaaS platform.
4. Configure any necessary environment variables or settings.
5. Deploy your web application to the PaaS platform.
6. Test your web application on the PaaS platform to ensure that it is functioning properly.

Step 4: Monitor and Scale the Web Application

1. Monitor the performance and usage of your web application on the PaaS platform.
2. Use the scaling features of the PaaS platform to adjust the resources allocated to your web application based on its usage.

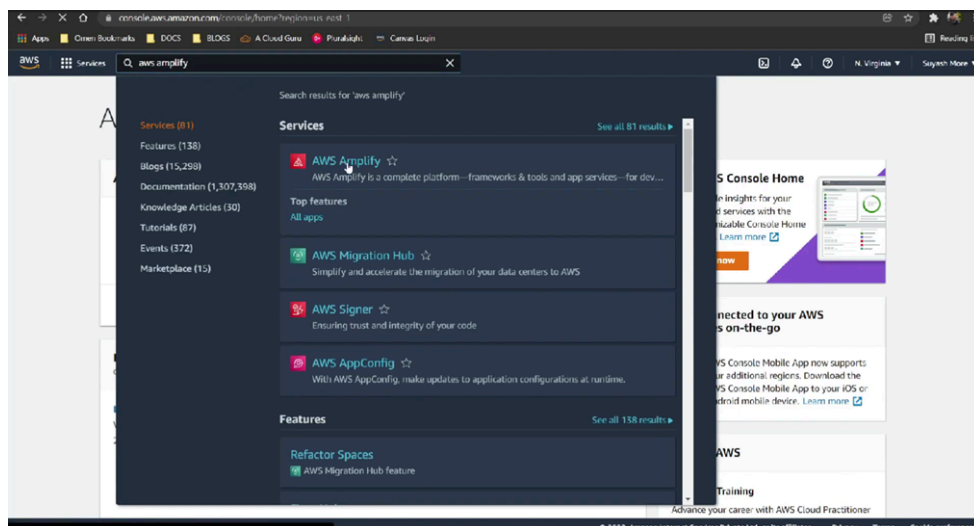
Procedure: To Deploy Web-Application on PAAS

Login to the AWS console

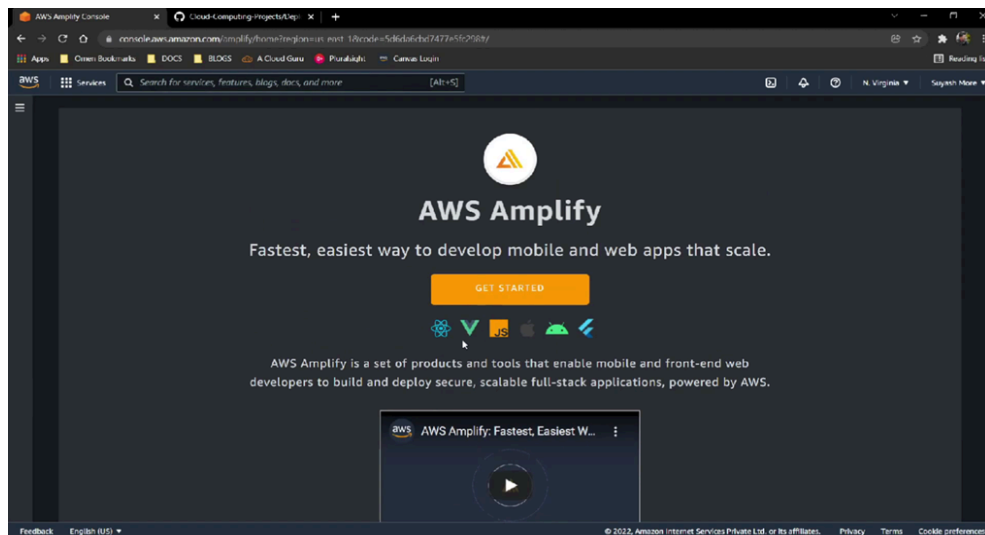


The image shows the AWS login page. At the top is the AWS logo. Below it is a section for login credentials. The first field is labeled "Account ID or alias" and contains a blurred input. The second field is labeled "IAM user name" and contains the text "udf". The third field is labeled "Password" and contains a series of dots. Below the password field is a blue "Sign In" button. At the bottom of the form are two links: "Sign-in using root account credentials" and "Forgot password?".

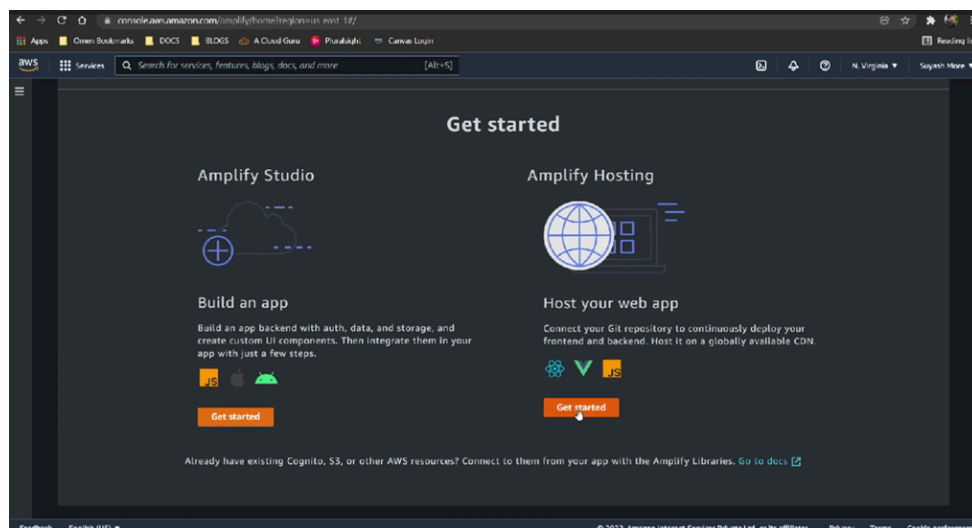
Find AWS Amplify in the services.



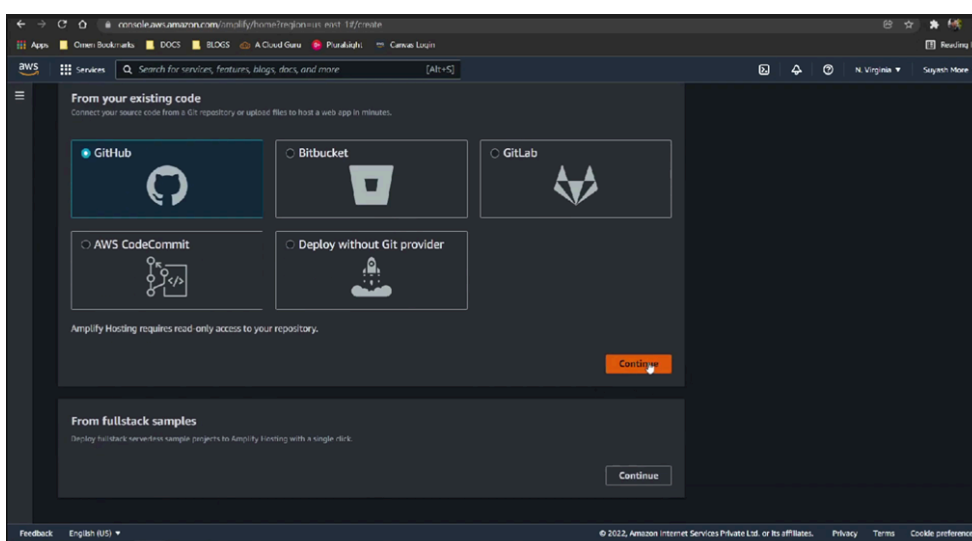
Get Started with Amplify service.



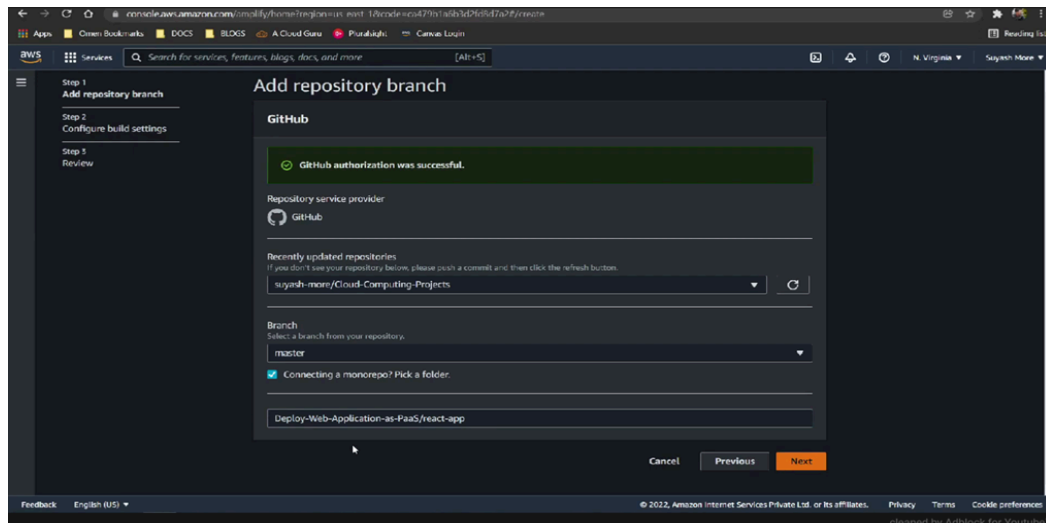
Click on Host a Web App.



Then choose to launch it with Git-hub and authenticate your GitHub account for the same.

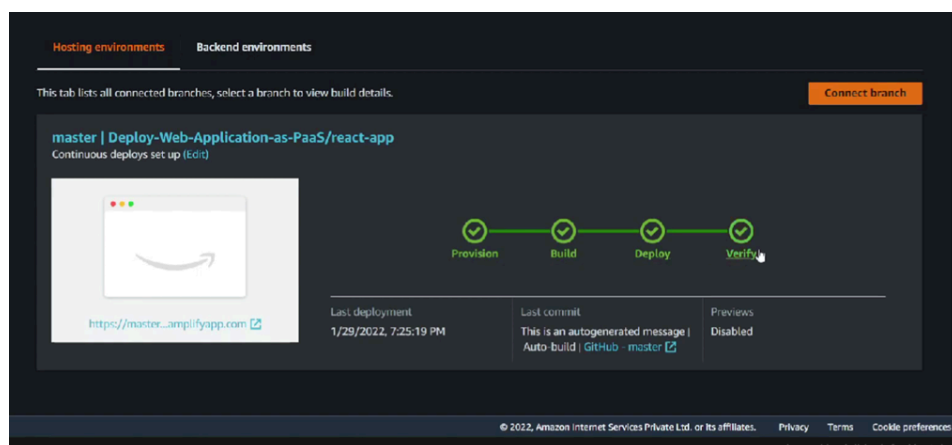


After that choose the Repository containing your source code (subfolder if needed)



After 4-Pipelines Launch the application with the default configurations provided by [AWSAmplify]

(<https://aws.amazon.com/amplify/#:~:text=AWS%20Amplify%20is%20a%20set,as%20your%20use%20cases%20evolve.>)



Configurations may be different on the type of framework/technology you are launching your application.

Conclusion:

We have successfully deployed our Web-Application on Amazon-Amplify which is one of the popular Platform as a service.

Assignment No. 7

Title :

Design and develop custom Application (Mini Project) using Salesforce Cloud.

Objective:

1. To learn about project development.
2. To learn how to build an application as a product.
3. To learn the basics of Salesforce Cloud development.
4. To create a custom application on Salesforce Lightning platform

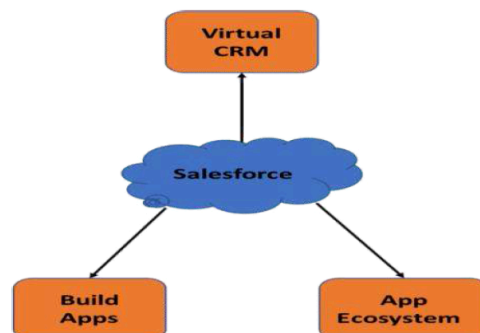
Requirements:

1. Salesforce Cloud development tool.
2. OS - Windows / Ubuntu, Google Chrome

Theory:

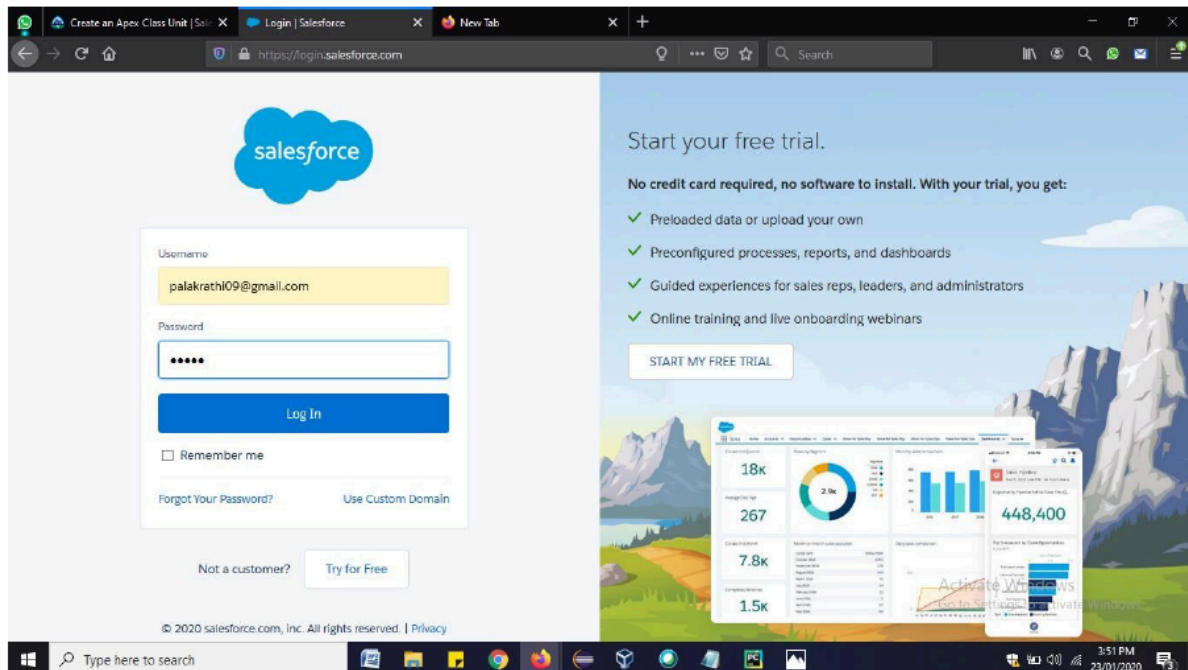
1. Salesforce: Salesforce is a cloud computing service as a software (SaaS) company that specializes in customer relationship management (CRM). Salesforce's services allow businesses to use cloud technology to better connect with customers, partners, and potential customers. The software has become the number one for customer success and helps businesses track customer activity, market to customers, and many more services. Salesforce is a customer relationship management solution that brings companies and customers together. It is one integrated CRM platform that gives all your departments — including marketing, sales, commerce, and service — a single, shared view of every customer.

2. Salesforce Lightning Experience: Salesforce Lightning Experience is simply referred to as “Lightning”. When working with lightning, we will learn about different Salesforce lightning topics like Lightning Login, Lightning App Builder, Lightning for Outlook, Salesforce Lightning Components, Lightning Sync, and many more. Some are applicable in Lightning Experience only, but others will work in both Lightning Experience and older Classic user interfaces. Lightning Sync syncs your user contacts and events between your email server and Salesforce.

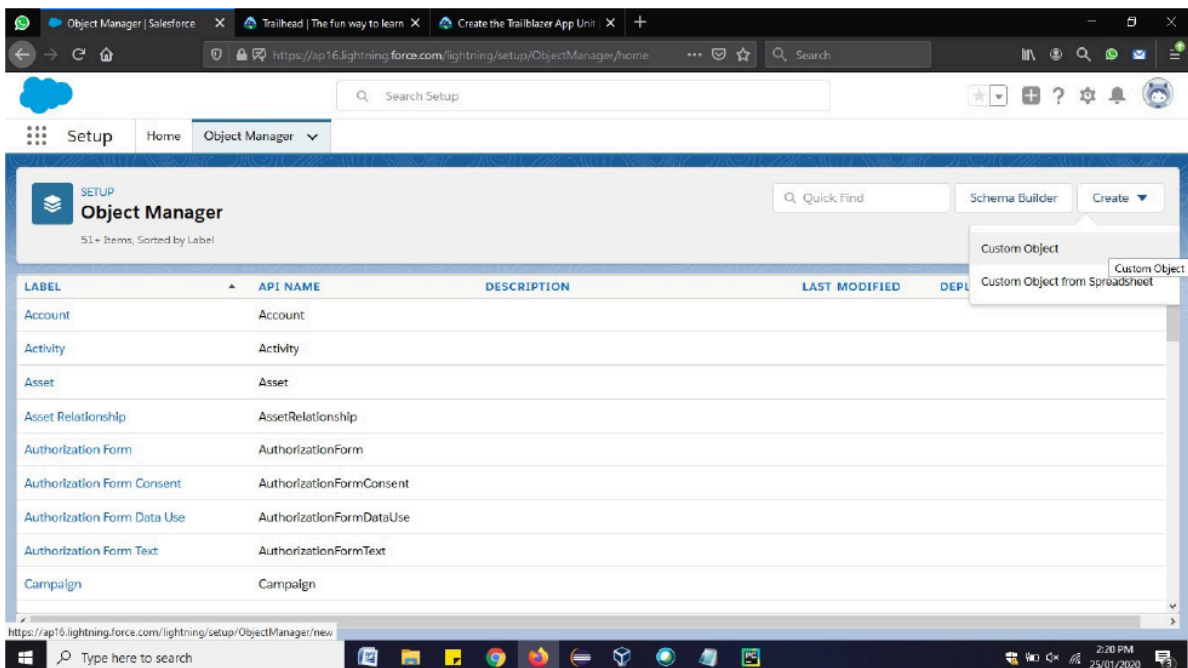


Application Development Steps:

Step-1: Log into the Salesforce Developer account.



Step 2: Switch to the Lightning mode by clicking on Lightning Experience. Open the Salesforce Lightning platform and click on Object Manager => Create => Custom Object.



Step 3: Fill in the required fields and under Optional Features, select **Allow Reports** and **Allow Activities**. Click **Save**.

Setup | Home | Object Manager

New Custom Object

Permissions for this object are disabled for all profiles by default. You can enable object permissions in permission sets or by editing custom profiles. [Tell me more!](#) [Don't show this message again](#)

Custom Object Definition Edit Save Save & New Cancel

Custom Object Information Required Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label Example: Account

Plural Label Example: Accounts

Starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name Example: Account Name

Date Type

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☐ Track Field History
- ☐ Allow in Chatter Groups

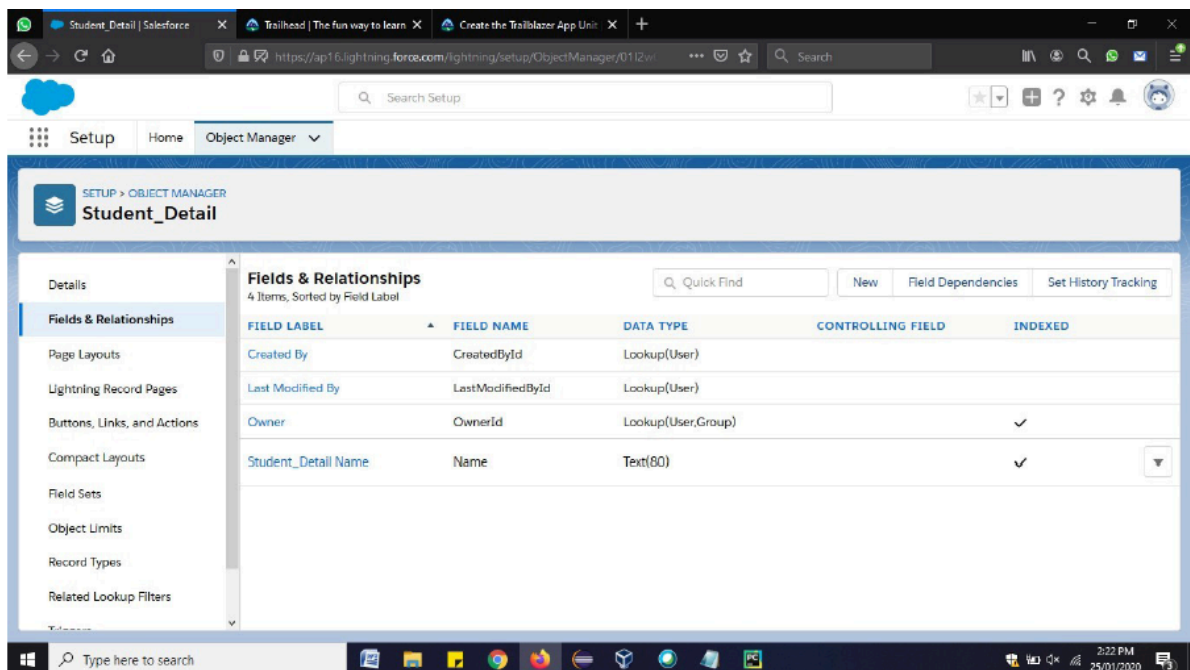
Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more](#)

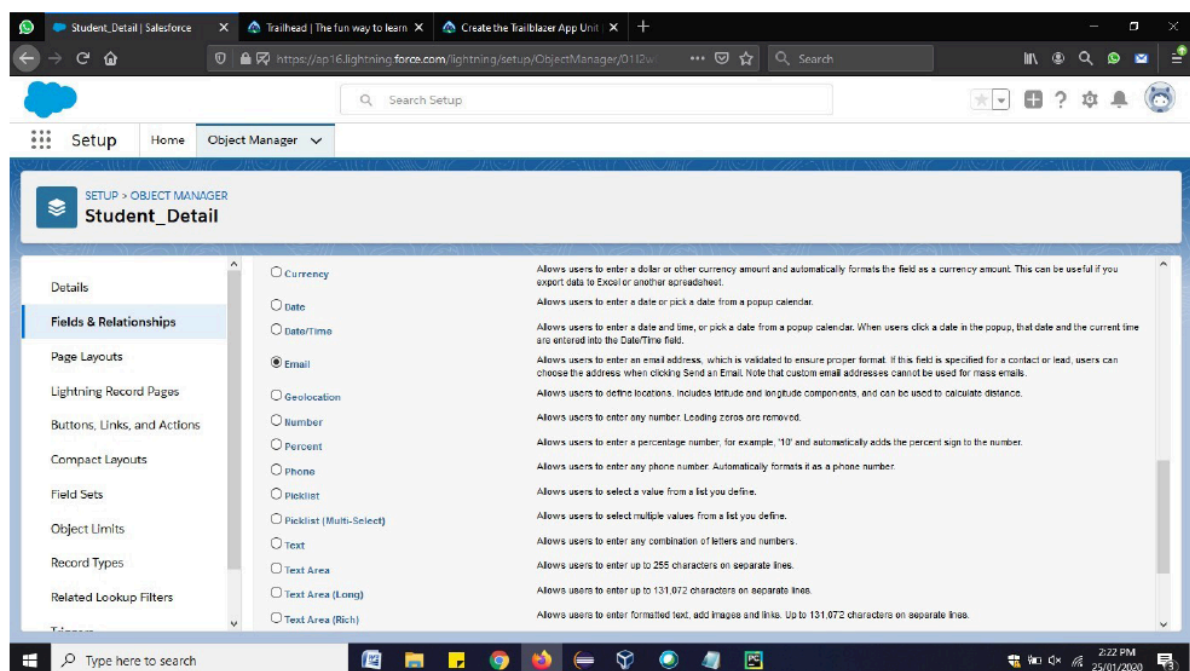
- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

Deployment Status [What is this?](#)

Step 4: Now, Click on Fields & Relations => New.



Step 5: Then select option “Email” and click Next-> Next -> Save.



Step 6: Similarly, repeat steps 4 and 5 to add more fields like Phone, Date of Birth. This is how the custom object will have the various fields.

The screenshot shows the Salesforce Setup interface for the 'Student_Detail' custom object. The 'Fields & Relationships' section is active, displaying a table of 6 fields. The table has columns for FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The fields listed are Contact, Created By, EmailID, Last Modified By, Owner, and Student_Detail Name.

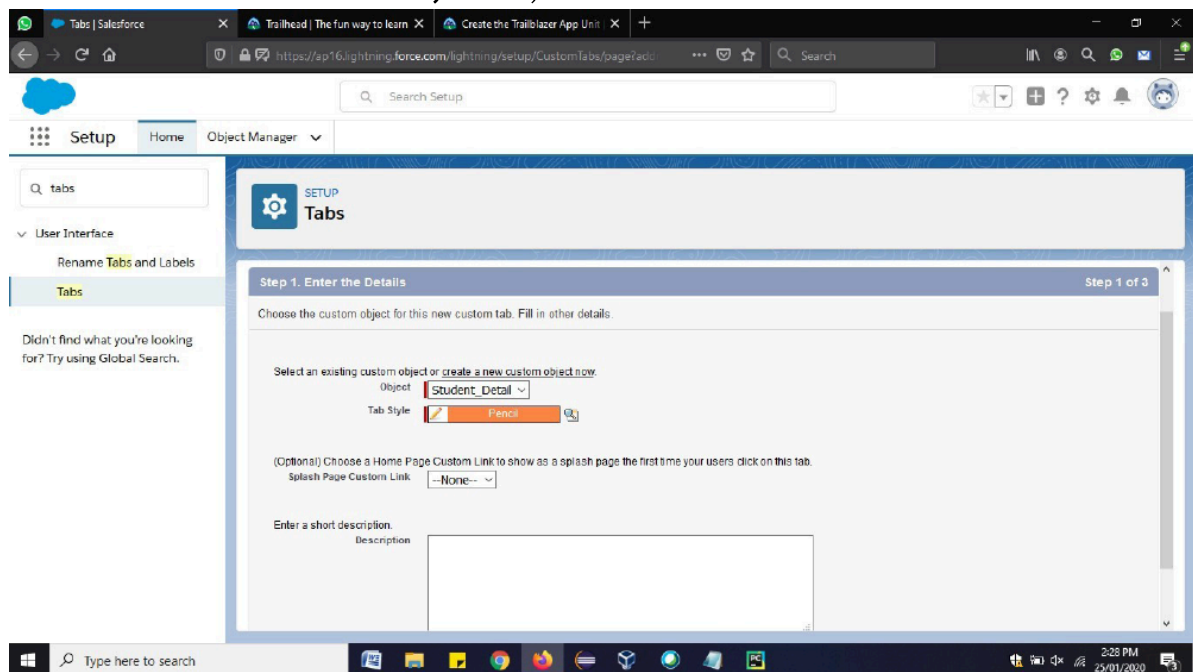
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Contact	Contact__c	Phone		
Created By	CreatedById	Lookup(User)		
EmailID	EmailID__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Student_Detail Name	Name	Text(80)		✓

Step 7: Now go to Home => search for “Tabs”. Click on New.

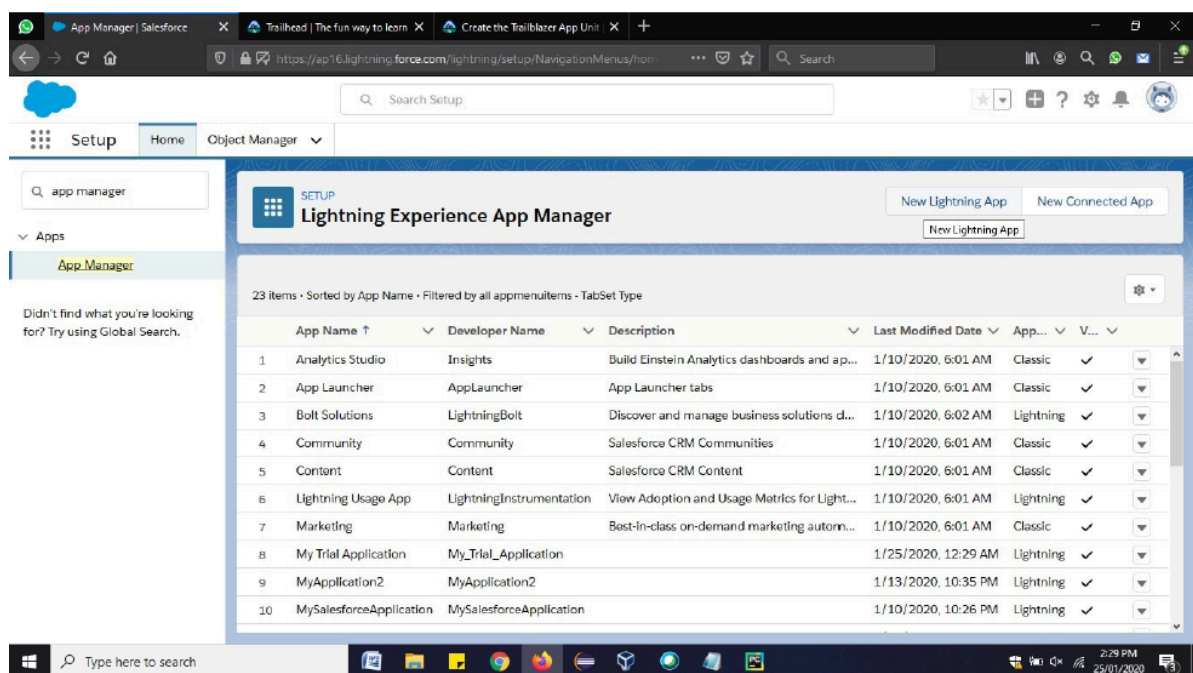
The screenshot shows the Salesforce Setup interface for the 'Tabs' section. The 'Custom Tabs' section is active, displaying a table of custom object tabs. The table has columns for Action, Label, Tab Style, and Description. The tabs listed are Comments, Names, Student_Names, StudentDetails, and xyz.

Action	Label	Tab Style	Description
Edit Del	Comments	Guitar	
Edit Del	Names	Pen	Enter your name
Edit Del	Student_Names	Star	
Edit Del	StudentDetails	Gears	
Edit Del	xyz	Cell phone	

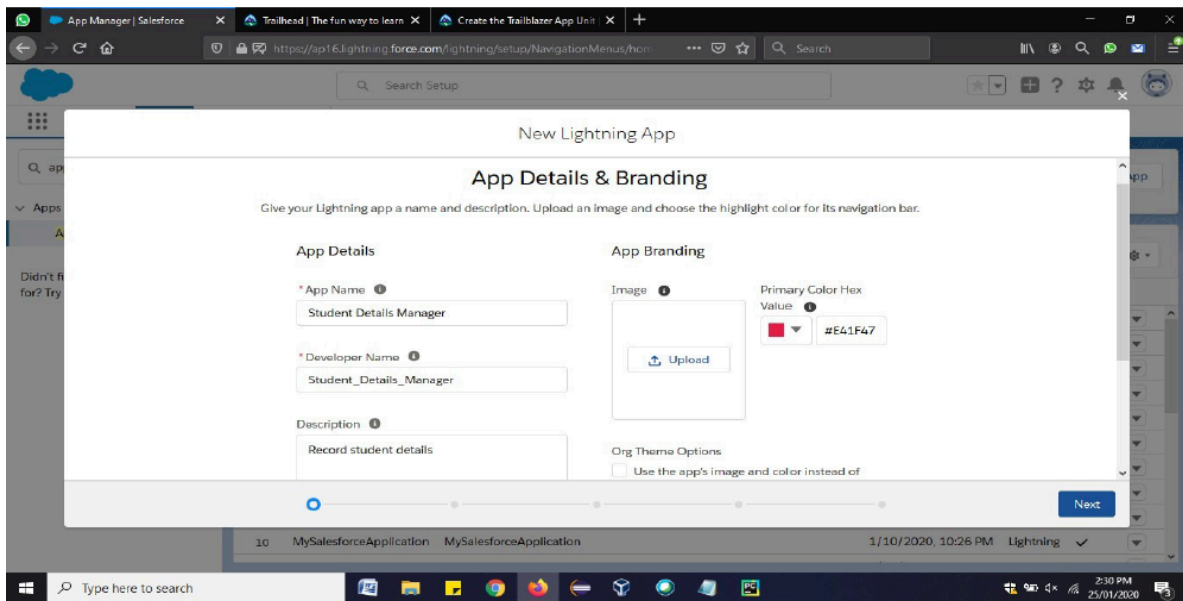
Step 8: Enter the Object name and select any icon for tab style. Leave all defaults as it is. Click **Next**, **Next**, and **Save**.



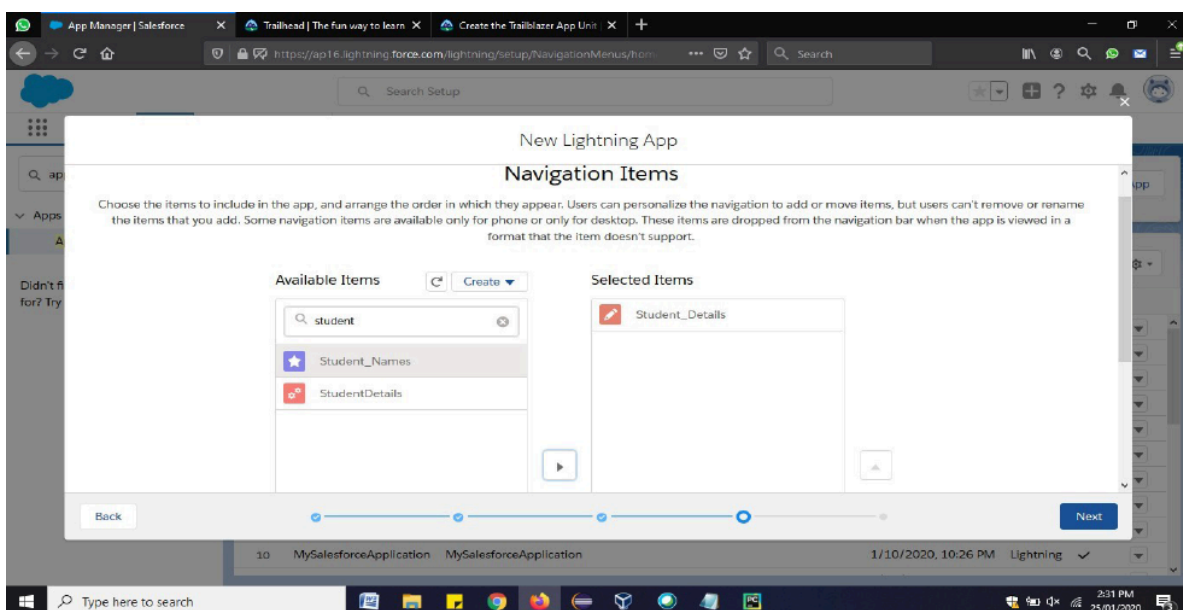
Step 9: In Setup, click **Home**. Enter “App Manager” in Quick Find and select **App Manager**. Click **New Lightning App**.



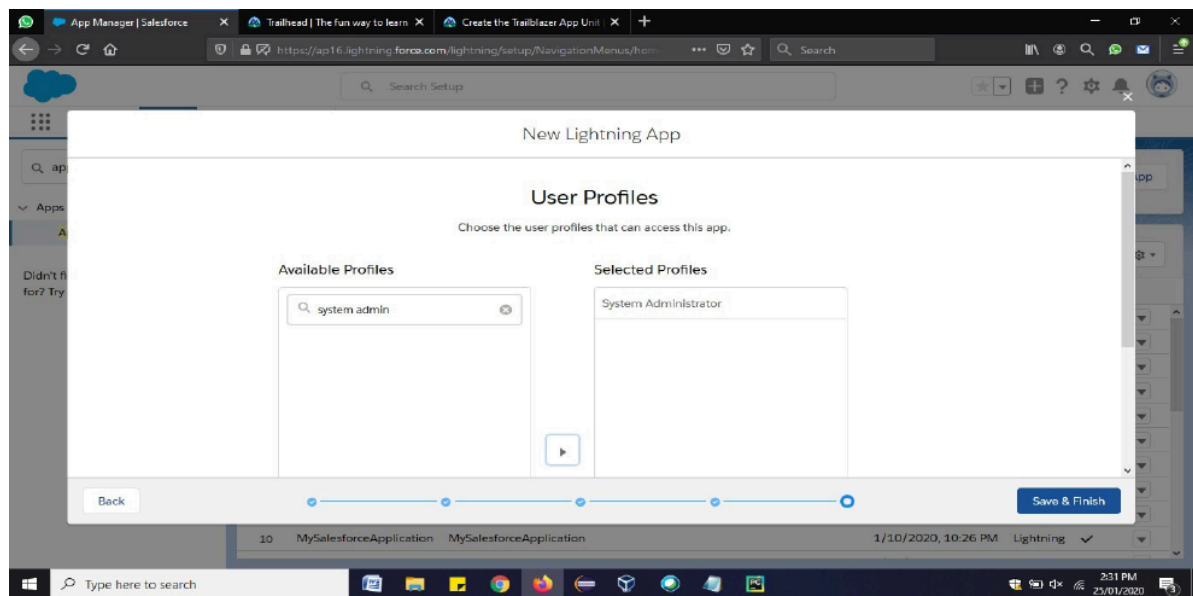
Step 10: Define the new Lightning app as follows:
App Name: Student Details Manager
Developer Name: Student_Details_Manager. Click **Next**.



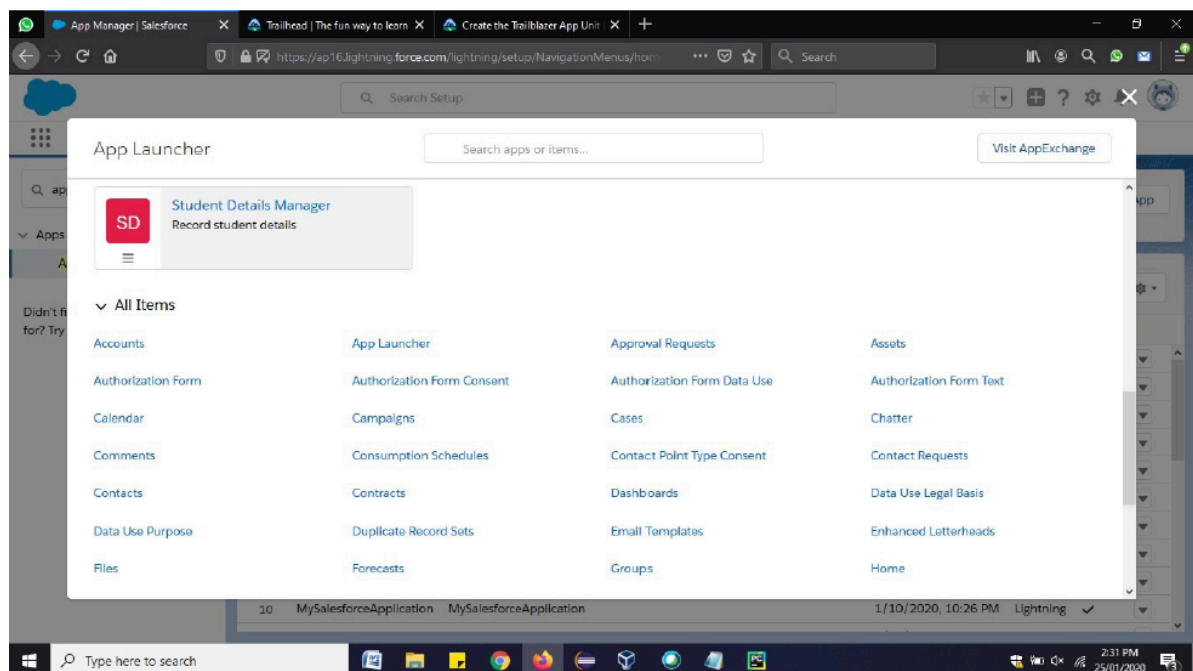
Step 11: On the App Options screen, leave the defaults as is and click **Next**. On the Utility Items screen, leave the defaults as is and click **Next**. On the Navigation Items screen, select **Student_Detail** and move them to the Selected Items box. Then click **Next**.

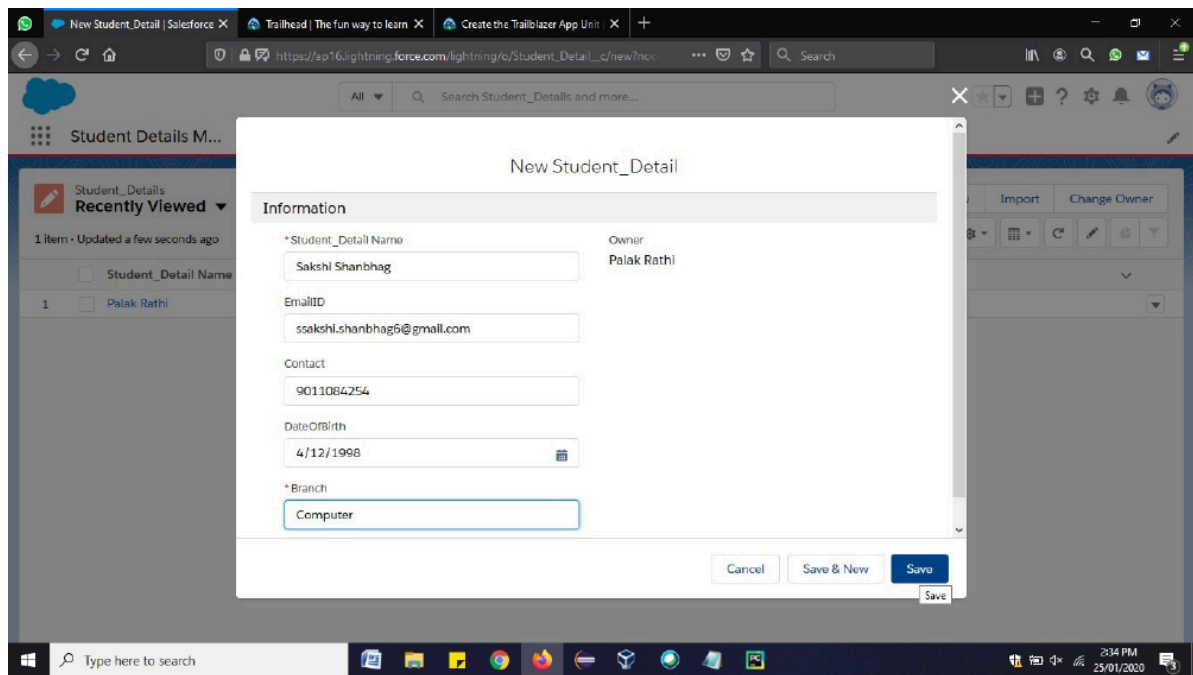


Step 12: On the Assign to User Profiles screen, select **System Administrator** and move it to Selected Profiles. Then click **Save & Finish**.

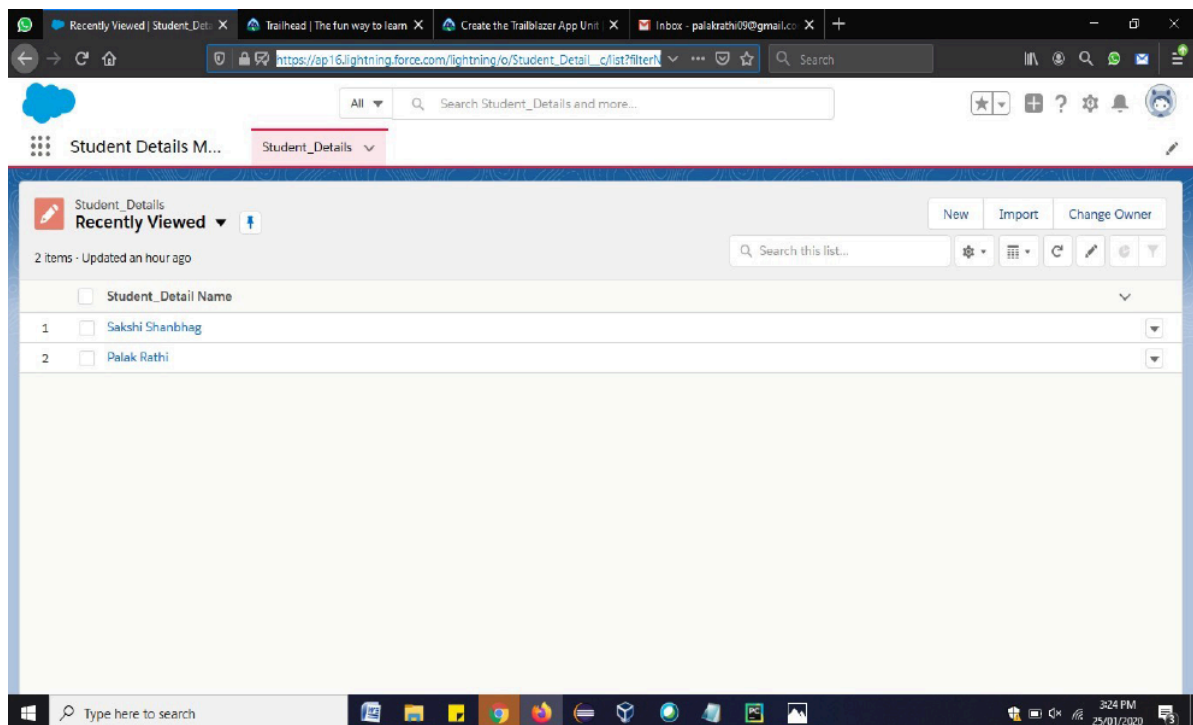


Step 13: Click on App launcher and Open the custom application created.

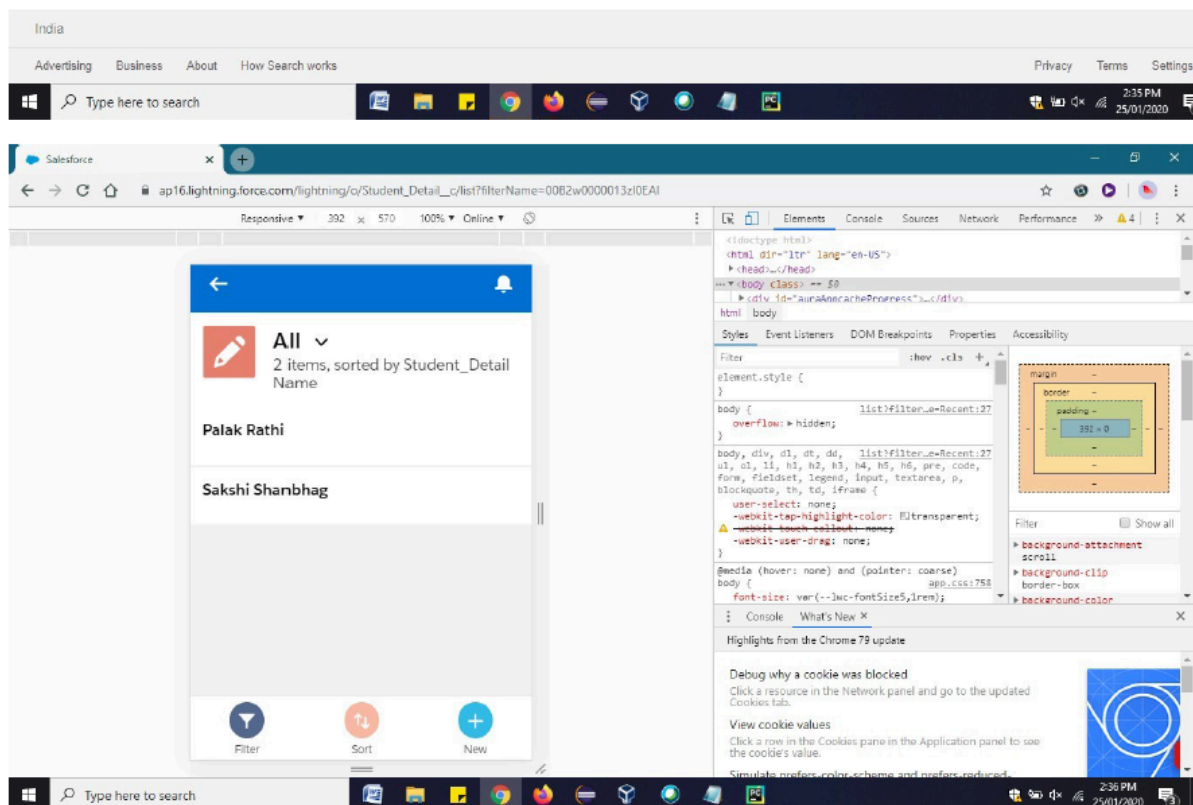
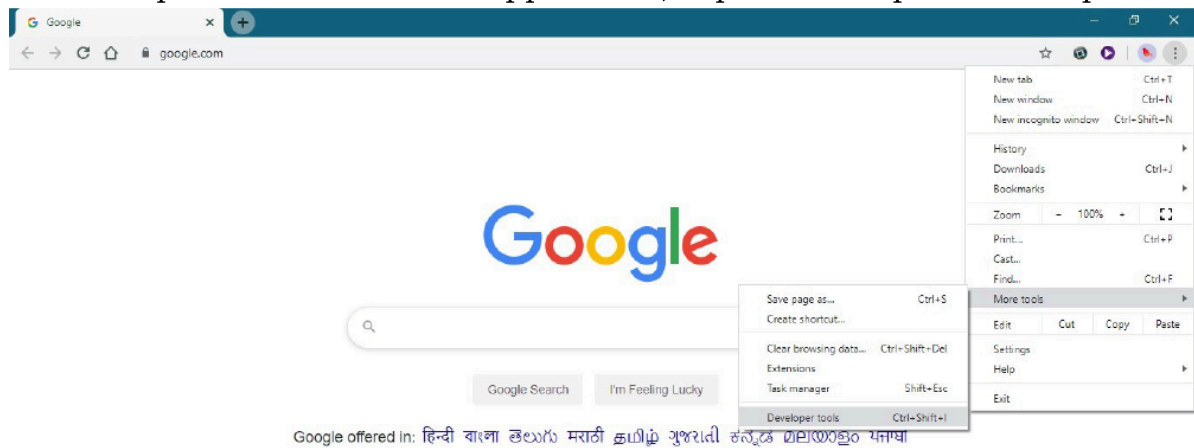


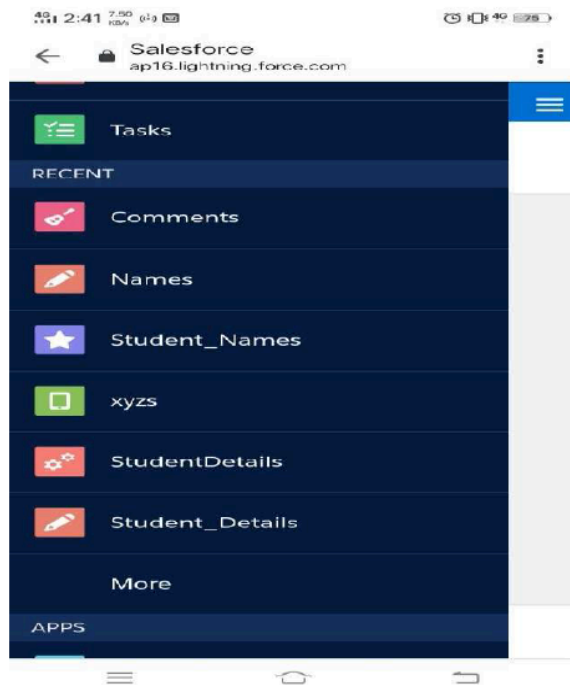


Step 14: Click on Student_Details => New => fill the specified details and copy the URL.

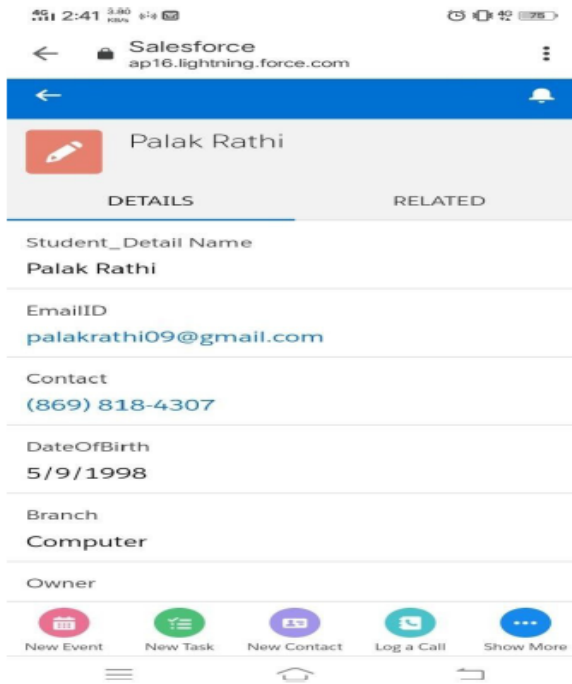


Step 15: Open Google Chrome new Tab => More Tools => Developer Tools and then paste the URL of the application, copied in the previous step.





This screenshot shows the 'New Student_Detail' form in the Salesforce mobile app. The status bar at the top shows the time as 2:41 and battery level at 76%. The app header displays the Salesforce logo and the URL 'ap16.lightning.force.com'. The form has a title bar with 'Cancel', 'New Student_Detail', and 'Save' buttons. The form is titled 'Information' and contains several input fields: 'Student_Detail Name' (marked with an asterisk), 'EmailID', 'Contact', 'DateOfBirth', and 'Branch' (marked with an asterisk). Below these fields, the 'Owner' is listed as 'Palak Rath'. The bottom of the screen shows the standard Android navigation bar with back, home, and recent apps icons.



Conclusion: We have learnt to create custom application using salesforce Lightning platform.

Assignment No. 8

Title :

Design an Assignment to retrieve, verify, and store user credentials using Firebase Authentication, the Google App Engine standard environment, and Google Cloud Datastore

Objective:

1. To learn about different cloud platforms.
2. To understand how we can use these platforms for application development and deployment.

Case Study-1

1. What is data storage in the private cloud?

Private cloud storage is a type of storage mechanism that stores an organization's data at in-house storage servers by implementing cloud computing and storage technology. Private cloud storage is similar to public cloud storage in that it provides the usability, scalability and flexibility of the storage architecture. But unlike public cloud storage, it is not publicly accessible and is owned by a single organization and its authorized external partners.

Private cloud storage is also known as internal cloud storage. Private cloud storage works much like public cloud storage and implements storage virtualization across an organization, providing a centralized storage infrastructure that can only be accessed by the authorized nodes.

Private cloud storage operates by installing a data center, which houses a series of storage clusters that are integrated with a storage virtualization application. Administrative policies and a management console provide access to the different storage nodes and applications within the organization's network. The applications or nodes access the private storage through file access and data retrieving protocols, while the automated storage administrator application allocates storage capacity to them on run time.

Private cloud storage has a multitenant architecture, where a single storage array can house storage space to multiple applications, nodes or departments.

2. Examples of data stored on cloud (success stories)

a. IT Priorities 2020: Covid-19 consolidates storage push to cloud

The emphasis on home and remote working has been felt in storage and backup, as in all areas of IT, with continued evolution towards the cloud and as-a-service storage and backup

b. Big storage meets cloud in the datacentre 'as-a-service' revolution

Cloud is mainstream but the datacentre is here to stay. This has resulted in a trend towards as-a-service in the datacentre where big storage array makers and the cloud giants meet.

3. Do you use cloud storage for these use cases yet?

We look at the use cases most suited to a quick transition to the cloud: backup, archiving, disaster recovery, file storage and cloud bursting – cloud storage's low-hanging fruit.

4. Cloud storage costs: How to get cloud storage bills under control?

Many organizations look to the cloud to cut storage budgets, but the potential costs are many and varied. So, what are the key ways to cut cloud storage costs?

5. Cloud storage 101: File, block and object storage in the cloud

We run the rule over file, block and object storage from the big three public cloud providers: AWS, Azure and GCP. We look at what's on offer and the use cases they are aimed at

6. Procedures/ ways to upload data to cloud (are there any applications designed)?

Here's how to build a cloud-ready application, and the application architecture concepts you'll need to succeed. The process presented below represents a step-by-step approach that mixes traditional software development concepts and reviews what's new with the cloud. It is a collection of best practices, concepts, and procedures for success.

Design the application as a collection of services

Cloud applications are best deployed as a collection of cloud services, or APIs. You build up from the data to the services and then combine those services into composite services or complete composite applications.

This is service-based or service-oriented architecture, at its essence. While many understand the concepts, developers still have a tendency to create tightly coupled applications that focus on the user interface, rather than expose the underlying functions as services they can leverage independently.

When developing an application architecture for the cloud, you deal with complex distributed systems that can take advantage of loosely coupled applications built on many services that can also be decoupled from the data (see "Decouple the data" below). You can separate the application services physically, executing on the proper machine instances, and service/API managers and governance technology that provide services directories can help track the many services that make up your application.

Additional benefits may include service reuse from other applications or more coarse-grained services. You can break up applications into hundreds of underlying services that have value when used by other applications. In this way, you're not reinventing the wheel each time you build an application. Consider the example of a credit-check service that many

applications use. Combine these into a single service and the application becomes much more efficient.

Decouple the data

If you tightly couple the data to the application, it won't find a good home in the cloud. Private and public clouds are complex distributed systems that work best with application architectures that break out processing and data into separate components.

You decouple the data for the same reason you want to build the application out of services. Once decoupled, you have the option to store and process the data on any public or private cloud instance. For example, many enterprises insist that their data remain on local servers but want to take advantage of the commodity virtual machine instances within a public cloud.

You must consider performance. Database reads and writes across the open Internet can cause latency, and database communications may determine how close your data sits to the services and applications that need to leverage it.

Consider using caching systems. These provide additional database performance by locally storing commonly accessed data, thereby reducing all database read requests back to the physical database. Those are best built into the application, however, and they should be tested with the application data to determine how efficient the cache will be. Systems that are constantly reading new data don't benefit as much from database caches.

Consider communications between application components

Decoupling applications, both data and services, doesn't mean your application is properly architected for the cloud. Chatty application components that constantly communicate with each other will lower the performance of the overall application, given that they're typically distributed over a network or the open Internet, where tolerance for high latency is desirable.

Focus on designing applications that optimize communications between application components. For example, combine communications into a single stream of data or a group of messages, rather than constantly communicating as if the application components reside on a single platform.

Model and design for performance and scaling

Extend considerations around how application components communicate to include overall performance as well. This includes understanding how the application will scale under an increasing load.

Designing for performance means first building a model that represents how the application behaves under an increasing load. If 1,000 or more users log on at the same time, how will the application handle the increased traffic on the network, the increased load on the application servers, and the load placed on the back-end databases? You need to

understand how application components handle the load as the number of users increases to 1,000 or more users.

This example might increase the load on the application servers by 80 percent, the load on the network by 10 percent, and the load on the database by 40 percent. Given that, adding 1,000 more users will likely saturate the application servers you've provisioned, and you'll need to spin up more application server instances. The network capacity might remain the same, but the number of database instances may have to increase to handle any additional load.

Armed with this model, you can figure out how best to scale the application by automatically spinning up resource instances that are needed. In some cases, cloud service providers offer auto-scaling capabilities, where provisioning occurs automatically. The most efficient path, however, lies in understanding the application's workload profile and defining the path to scaling the application, as well as putting mechanisms in place to ensure that it will, indeed, scale.

Finally, monitor overall application performance using application-aware performance monitoring tools, and create interfaces within the application to better enable performance monitoring. How the applicable provisions and de-provisions resources should be innate to the application as well.

Make security systemic within the application

For most people who build applications, security is typically an afterthought. When hosting an application in the cloud, however, security should be a high priority. Your cloud-based application architecture should make security systemic to the application—it should be designed and built into the application architecture.

Pick a security approach and technology prior to building your application that will be effective for the type of application you're running and that will address any compliance or other data-level security issues. If you're in healthcare, for example, you must consider personally identifiable information as well as the Health Insurance Portability and Accountability Act (HIPAA) in the US. You'll need to store data in a certain way, on clouds that are HIPAA-compliant. Moreover, the application will need to handle sensitive data in specified ways, with required levels of security, such as encryption.

Generally speaking, cloud-based applications should leverage identity and access management (IAM). Enterprises that develop mature IAM capabilities can reduce their security costs and, more importantly, become significantly more agile at configuring security for cloud-based applications. Indeed, IAM will be a part of more than 50 percent of existing applications that migrate to the public cloud and nearly 90 percent of new applications built on clouds.

What's more, the use of IAM within cloud application deployments will backfill into the enterprise, as these organizations modernize security approaches and technologies to align with the use of public clouds. In many cases, IAM will be provided as a service to the enterprise. This

concept of cloud-delivered IAM quickly leads to the concept of centralized identity management. As you build more cloud-based applications using IAM, each application should become significantly more secure and more cost effective.

Your core objective is to design security into the application and take advantage of the native features of both the cloud and the IAM system you use. However, each application has its own requirements based upon the needs of the business, and security always differs from one enterprise to another.

Building a cloud-ready application architecture requires that you pay attention to a few new things, but many of the traditional concepts are still important, such as sound design, testing, and learning from your mistakes. Most developers who deploy applications on private or public cloud platforms will make some blunders, but as long as they recognize, correct, and learn from those mistakes, they'll be well on their way to finding a more effective path to building applications in the cloud.

Understand that approaches such as service orientation should be given priority, even if it means longer initial application development lifecycles and bigger budgets. Even though you'll pay more for application development in the cloud than you did for traditional application development, the investment in services pays huge dividends year in and year out. It's a smart investment.

7. Security algorithm in place for data security in cloud (list , brief differences , latest standard followed)

Cloud computing security or, more simply, cloud security refers to a broad set of policies, technologies, and controls deployed to protect data, applications, and the associated infrastructure of cloud computing. It is a sub-domain of computer security, network security, and, more broadly, information security.

Cloud computing security is the set of control-based technologies and policies designed to adhere to regulatory compliance rules and protect information, data applications and infrastructure associated with cloud computing use.

Because of the cloud's very nature as a shared resource, identity management, privacy and access control are of particular concern. With more organizations using cloud computing and associated cloud providers for data operations, proper security in these and other potentially vulnerable areas have become a priority for organizations contracting with a cloud computing provider.

Cloud computing security processes should address the security controls the cloud provider will incorporate to maintain the customer's data security, privacy and compliance with necessary regulations. The processes will also likely include a business continuity and data backup plan in the case of a cloud security breach.

The following are 10 security-as-a-service categories are :

1. Identity and Access Management should provide controls for assured identities and access management. Identity and access management includes people, processes and systems that are used to manage access to enterprise resources by assuring the identity of an entity is verified and is granted the correct level of access based on this assured identity. Audit logs of activity such as successful and failed authentication and access attempts should be kept by the application/solution.
2. Data Loss Prevention is the monitoring, protecting and verifying the security of data at rest, in motion and in use in the cloud and on-premises. Data loss prevention services offer protection of data usually by running as some sort of client on desktops/servers and running rules around what can be done. Within the cloud, data loss prevention services could be offered as something that is provided as part of the build, such that all servers built for that client get the data loss prevention software installed with an agreed set of rules deployed.
3. Web Security is real-time protection offered either on-premise through software/appliance installation or via the cloud by proxying or redirecting web traffic to the cloud provider. This provides an added layer of protection on top of things like AV to prevent malware from entering the enterprise via activities such as web browsing. Policy rules around the types of web access and the times this is acceptable also can be enforced via these web security technologies.
4. E-mail Security should provide control over inbound and outbound e-mail, thereby protecting the organization from phishing and malicious attachments, enforcing corporate policies such as acceptable use and spam and providing business continuity options. The solution should allow for policy-based encryption of e-mails as well as integrating with various e-mail server offerings. Digital signatures enabling identification and non-repudiation are features of many cloud e-mail security solutions.
5. Security Assessments are third-party audits of cloud services or assessments of on-premises systems based on industry standards. Traditional security assessments for infrastructure and applications and compliance audits are well defined and supported by multiple standards such as NIST, ISO and CIS. A relatively mature toolset exists, and a number of tools have been implemented using the SaaS delivery model. In the SaaS delivery model, subscribers get the typical benefits of this cloud computing variant elasticity, negligible setup time, low administration overhead and pay-per-use with low initial investments.
6. Intrusion Management is the process of using pattern recognition to detect and react to statistically unusual events. This may include reconfiguring system components in real time to stop/prevent an intrusion. The methods of intrusion detection, prevention and response in physical environments are mature; however, the growth of virtualization and massive multi-tenancy is creating new targets for intrusion and raises many questions about the implementation of the same protection in cloud environments.

7. Security Information and Event Management systems accept log and event information. This information is then correlated and analyzed to provide real-time reporting and alerting on incidents/events that may require intervention. The logs are likely to be kept in a manner that prevents tampering to enable their use as evidence in any investigations.

8. Encryption systems typically consist of algorithms that are computationally difficult or infeasible to break, along with the processes and procedures to manage encryption and decryption, hashing, digital signatures, certificate generation and renewal and key exchange.

9. Business Continuity and Disaster Recovery are the measures designed and implemented to ensure operational resiliency in the event of any service interruptions. Business continuity and disaster recovery provides flexible and reliable failover for required services in the event of any service interruptions, including those caused by natural or man-made disasters or disruptions. Cloud-centric business continuity and disaster recovery makes use of the cloud's flexibility to minimize cost and maximize benefits.

10. Network Security consists of security services that allocate access, distribute, monitor and protect the underlying resource services. Architecturally, network security provides services that address security controls at the network in aggregate or specifically addressed at the individual network of each underlying resource.

In a cloud/virtual environment, network security is likely to be provided by virtual devices alongside traditional physical devices.

8. Diagrammatic representation of Data storage security in private cloud

Conclusion:

1. Here we studied Data Storage related to the private cloud.
2. Learned about the techniques used to store data in the private cloud.
3. Security Algorithm and diagrammatic representation of security implemented in the private cloud.